

12/4/2025

DERMATOLOGICAL PHARMACOLOGY

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RECOMMENDED RESOURCES FOR READING

2

- Katzung: Basic and Clinical Pharmacology
 - Ch 61: Dermatologic Pharmacology
 - bread and butter of topic summary; comprehensive lists of drug classes with derm application
- DiPiro's Pharmacotherapy: a Pathophysiological Approach focus on new drugs
 - Section 16: Dermatologic Disorders
 - Chapter 113: Acne Vulgaris
 - Chapter 114: Psoriasis
 - Chapter 115: Atopic Dermatitis
 - Chapter 116: Alopecia
 - Chapter e117: Dermatologic Drug Reactions, Contact Dermatitis, and Common Skin Conditions
 - Chapter 156: Melanoma
 - do not read all these chapters unless you want to; recommended as supplemental ref



All these books can be found at: <https://libguides.tu.edu/COMbooks/pharmacology>

...but you don't have to take my word for it....

LECTURE OUTLINE

Drug lists (on your own)

Your skin is a living thing

PK review

Drug-induced derm emergencies

Overview of treatments for common conditions

acne

psoriasis

eczema/atopic dermatitis

neoplasms

Retinoids

Photochemotherapy

Photoprotection: sunscreens

DRUG **CLASSES** TO KNOW

• **Antibiotics**

- cell wall synthesis inhibitors:
 - bacitracin, gramicidin
- protein synthesis inhibitors:
 - neomycin, gentamycin, tetracyclines
 - **clindamycin, erythromycin (DOC in acne)**
 - **ozenoxacin, muciprocin, retapamulin**
 - **effective against MRSA**
- polymyxin B sulfate
- metronidazole (rosacea)
- ivermectin (rosacea, mites)
- sodium sulfacetamide (acne)
- dapsone (acne)

• **Antifungals: topical**

- azoles, e.g., **miconazole**, clotrimazole, efinaconazole, econazole, ketoconazole, luliconazole, oxiconazole, sertaconazole, sulconazole
- ciclopirox olamine
- naftifine
- terbinafine
- butenafine
- tolnaftate
- nystatin
- amphotericin B

• **Antifungals: oral**

- griseofulvin
- terbinafine
- **fluconazole**
- itraconazole

These drugs are not unique to dermatology: brief coverage only in these slides.

DRUG **CLASSES** TO KNOW, CONT'D

- **Antivirals**

- acyclovir
- valacyclovir
- penciclovir
- famciclovir

- **antiproliferatives and keratinolytic drugs**

- 5-flourouracil
- methotrexate
- salicylic acid
- azaleic acid (note multiple MOA)

- **anti-ectoparasitics**

- **insecticides/pediculicide**

- permethrin
- spinosad
- ivermectin
 - also antihelminthic
- lindane
 - note ovicide activity also
- crotamiton
- malathion

These drugs are not unique to dermatology: brief coverage only in these slides.

- **immunomodulators**

psoriasis, eczema

- **TNF- α inhibitors**, e.g.,

- adalimumab
- etanercept
- infliximab

- **cytokine inhibitors**, e.g.,

- ixekizumab
- secukinamab

- **Toll-like receptor agonists**, e.g.,

- imiquimod (warts, carcinomas)

- **calcineurin inhibitors**, e.g.,

- tacrolimus, pimecrolimus

- **anti-inflammatories**

- steroid, e.g., corticosteroids
- non-steroid, e.g., crisaborole

- **PDE4 inhibitors**

- apremilast (psoriasis)

DRUGS TO KNOW: DERM-SPECIFIC

- **pigmentation and Vit D modulators**

- hydroquinone, mequinol (hyperpigmentation)
- monobenzone (hyperpigmentation)
 - irreversible depigmentation
- psoralens (vitiligo)
 - trioxsalen
 - methoxsalen
- calcipotriene, calcitriol (psoriasis)

- **retinoids**

- tretinoin
- isotretinoin
- retinoic acid
- adapalene
- tazarotene
- acitrecin

**These drugs
ARE unique to
dermatology so
more extensive
coverage here.**

- **photoblockers**

mineral

- TiO_2 , ZnO_2

- **photoabsorbers**

chemical

- homosalate

- octisalate

- avobenzone

- oxybenzone

- octocrylene

- octinoxate

- benzophenones

- oxybenzophenone

- dioxybenzone

- sulizobenzene

FYI: combo in Neutrogena Beach Defense SPF 70, the sunscreen in my house that I read the ingredient list of while prepping this lecture. I do not endorse or receive anything from Neutrogena 😊

STUDY GUIDE: WHAT TO FOCUS ON

- Know **CLASSES** of drugs used for common dermatological conditions
 - know some common examples, especially if preferred treatment/DOC
- Explain the most common clinical strategies for treating **acne, psoriasis, eczema/atopic dermatitis, and hyperpigmentation**
- Recognize **DERMATOLOGICAL EMERGENCIES** and how to address them
 - SJS, TEN, DRESS, purpura fulminans, AGEP, pyoderma gangrenosum
- Explain **RETINOIDS** MOA, indications, C/I, and SE
 - e.g., molecular mechanisms of action, which are C/I in pregnancy, retinoid-induced dermatitis and skin irritation
- Explain **PSORALEN** and **PHOTOCHEMOTHERAPY** MOA, indications, C/I, and SE
 - remember that psoralens don't work alone; they need to see the light (i.e., are light-activated)
- Explain how **SUNSCREENS** work
 - Physical/inorganic filtering and chemical/organic filtering agents
 - explain MOA with respect to UV wavelength – distinguish UVA and UVB damage to skin and agents targeting each range
 - explain SPF (it's an adjective, not a noun)

PHARMACOKINETIC CONSIDERATIONS OF THE SKIN

- **absorption**

- topical formulations
 - concept note: allows use of agents not indicated for systemic use!
- thickness of stratum corneum
- skin hydration
- occlusion of preparation

- **distribution**

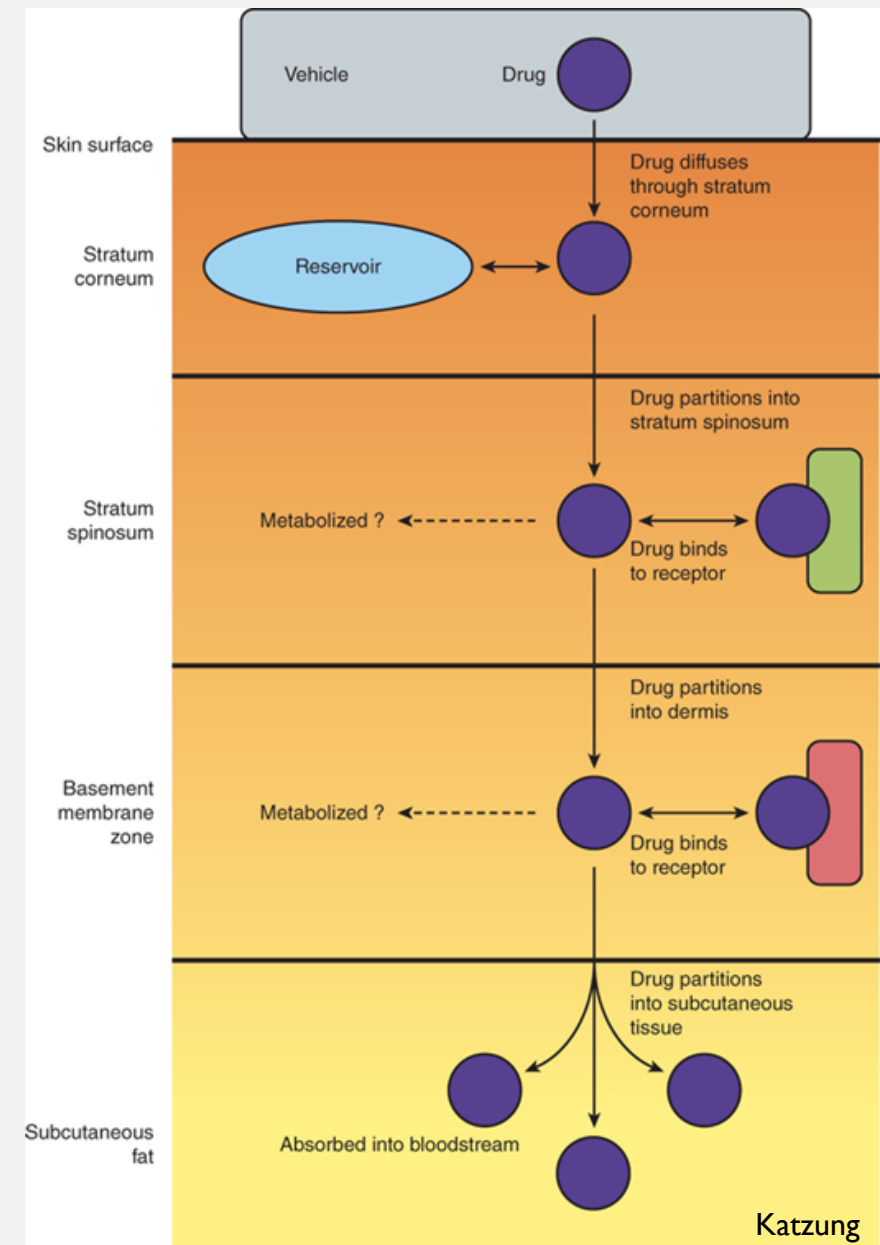
- partition coefficient of drug
 - lipophilicity
 - hydrophilicity

- **metabolism**

- catalytic activity present in **stratum spinosum**, **basement membrane**

- **elimination**

- degree of vascularization
 - to carry drug away from site
- subsequent systemic effects
 - including metabolism and excretion



LOCAL CUTANEOUS REACTIONS TO TOPICAL DRUGS

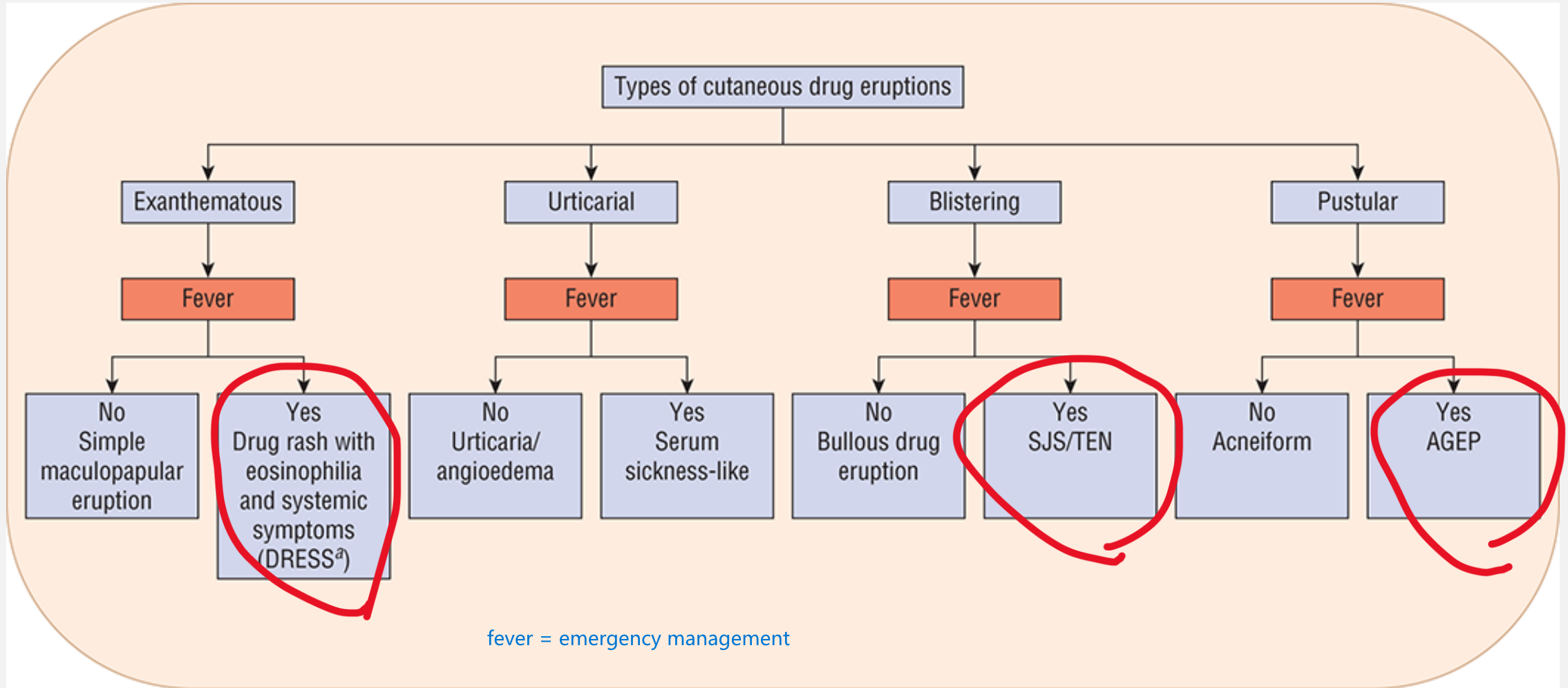
Reaction Type	Mechanism	Comment
Irritation	Nonallergic	Most common local reaction
Photoirritation	Nonallergic	Phototoxicity; usually requires UVA exposure
Allergic contact dermatitis	Allergic	Type IV delayed hypersensitivity
Photoallergic contact dermatitis	Allergic	Type IV delayed hypersensitivity; usually requires UVA exposure
Immunologic contact urticaria	Allergic	IgE-mediated type I immediate hypersensitivity ; may result in anaphylaxis
Nonimmunologic contact urticaria	Nonallergic	Most common contact urticaria; occurs without prior sensitization

dermatitis - over days

hive min-hrs



DERMATOLOGIC EMERGENCIES: REACTIONS TO MEDICATION



DERMATOLOGICAL EMERGENCIES TO KNOW

- **Stevens-Johnson syndrome (SJS)** and **toxic epidermal necrolysis (TEN)**: allergic response to certain drugs.

- **AN ACUTE LIFE-THREATENING EMERGENCY**

- common drug “triggers”: sulfonamides, penicillins, some anticonvulsants (hydantoins, carbamazepine, barbiturates, lamotrigine), NSAIDs, and allopurinol *ethosuximide*
- acute onset (7-14 days of drug exposure)
- generalized tender or painful bullous formation with accompanying systemic signs and symptoms, including fever, headache, and respiratory symptoms
- **rapid deterioration**, including extensive epidermal detachment and sloughing
- Treatment: **ICU**

- **Drug Rash with Eosinophilia & Systemic Symptoms (DRESS) Syndrome**

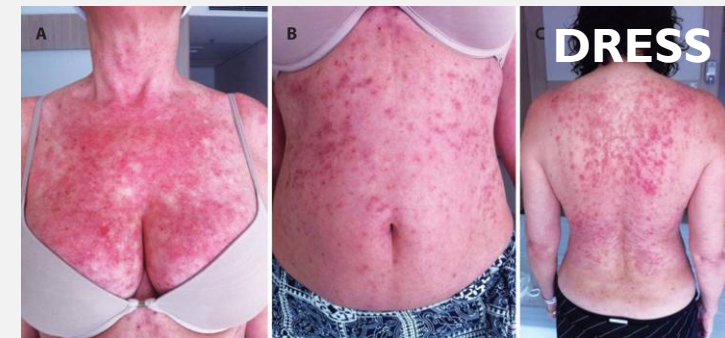
- common drugs: allopurinol, sulfonamides, some anticonvulsants (barbiturates, phenytoin, carbamazepine, lamotrigine), and dapsone
- Not infection; treat with prolonged **immunosuppressants**



<https://metro.co.uk/2021/06/02/black-woman-denied-help-as-white-doctors-didnt-recognise-symptoms-14699532/>

SJS rxn to **carbamazepine**; drs did not take drug history or recognize SJS

UptoDate



DERMATOLOGICAL EMERGENCIES TO KNOW

- Purpura Fulminans

- can be caused by meningococcal infection
- May develop disseminated intravascular coagulation (DIC)
- treat with **broad spectrum antibiotics**



UptoDate

- Acute Generalized Exanthematous Pustulosis (AGEP)

- common drugs: antibiotics, antifungals, , antimalarials, androgenic hormones, some anticonvulsants (e.g., ethosuximide), diltiazem, isoniazid, and lithium.
- Treat with topical CS or systemic **immunosuppressants**



UptoDate

- Pyoderma Gangrenosum

- autoinflammatory ulcerative skin disorder; Not infection or allergic reaction; treat with **immunosuppressants (e.g. CS)**



UptoDate

BOARDS-STYLE EXAMPLE

- A 7-year-old female patient presents to her pediatrician's office, brought by her parent. The child's teacher reported that the patient has periods of time where she stares at the wall and seems to "space out." These "space out" episodes last about 10 seconds and occur multiple times a day. During the episodes, she does not lose body tone or have violent movements of the limbs. The patient is not aware of these episodes while they occur and continues with whatever activity she was doing prior to the episode. The patient's medical history is otherwise unremarkable, and she has no known drug allergies. After appropriate testing, she is placed on an appropriate pharmacotherapy treatment regimen. Given the most likely prescribed medication for this condition, which of the following side effects would be the most appropriate for the physician to discuss with the mother?

ethosuximide - anticonvulsant

- A. extrapyramidal symptoms
- B. gingival hyperplasia
- C. pseudomembranous colitis
- D. respiratory depression
- ☒ E. Stevens-Johnson syndrome

Bottom Line: Typical absence seizures present with loss of consciousness without loss of body tone, lasting on the average of 10 seconds. Ethosuximide is the preferred treatment. As with other antiepileptics, a potentially serious side effect is Stevens-Johnson syndrome.

COMMON DERMATOLOGICAL CONDITIONS

- acne
 - treatments target oil production, gland occlusion, bacterial growth, inflammation
- psoriasis
 - treatments target autoimmune excess, cell proliferation
- eczema/atopic dermatitis
 - treatments target hypersensitivity
- neoplasms: SCC, BCC, melanoma, cutaneous T-cell lymphoma, others
 - treatments target cancer cell proliferation
- infection/infestation
 - treatments target microbes/parasites
- hyperpigmentation
 - treatment: melanin synthesis inhibitors, melanocyte disruption
- sunburn/photoaging UVB/UVA
 - treatment is supportive; preventive care is photoprotection

TREATMENT OF ACNE

Cutibacterium (formerly *Propionibacterium*) *acnes* proliferation:

- Benzoyl peroxide
- Topical and oral antibiotics
- Azelaic acid

Follicular hyperproliferation and abnormal desquamation:

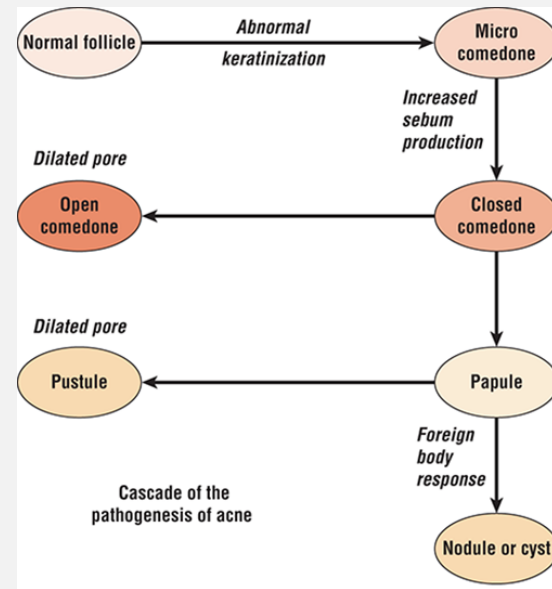
- Topical retinoids
- Oral isotretinoin
- Azelaic acid
- Salicylic acid

Increased sebum production:

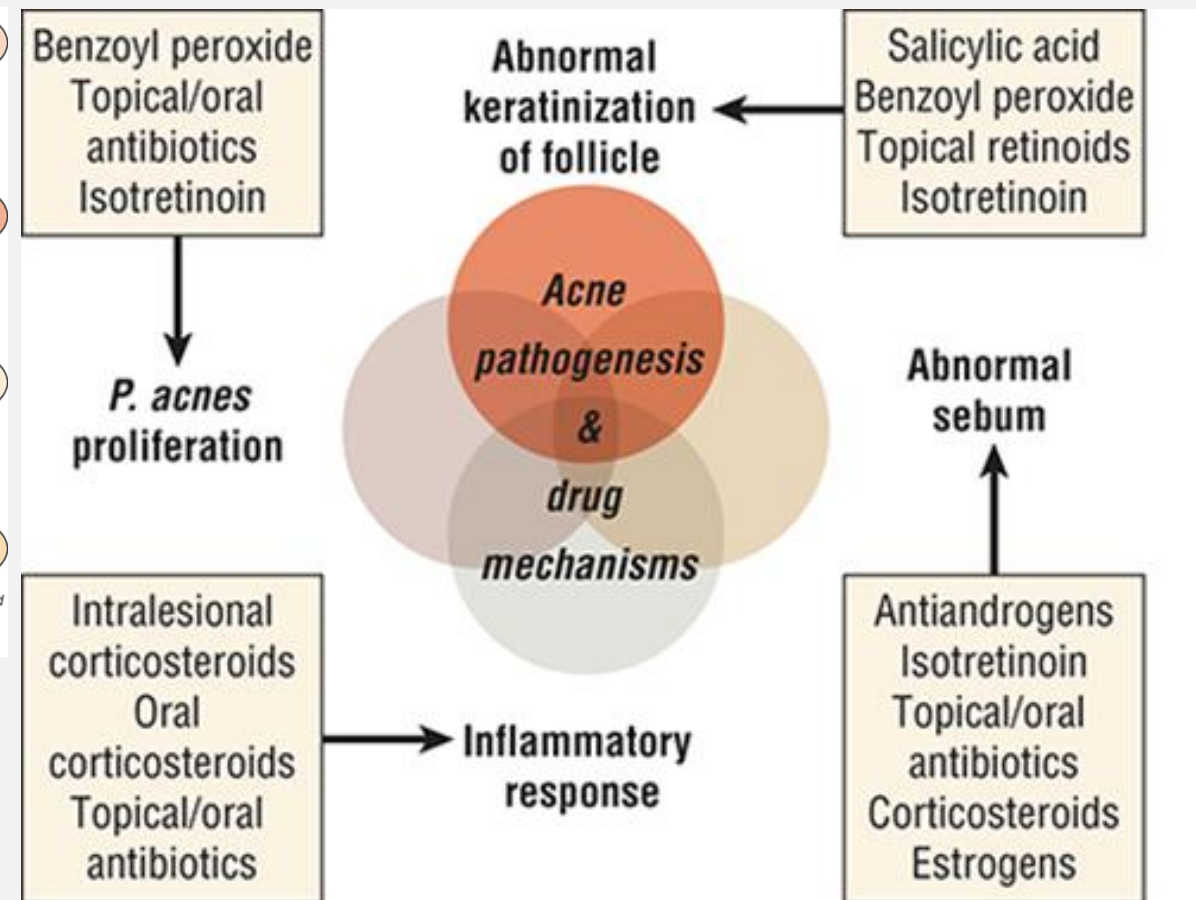
- Oral isotretinoin
- Oral contraceptives if hormonal
- Spiroglactone anti-androgens
- Clascoterone

Inflammation:

- Oral isotretinoin
- Oral tetracyclines
- Topical retinoids
- Azelaic acid
- Topical dapsone also for mycobacterium infections like TB



Source: JT DiPiro, GC Yee, LM Posey, ST Haines, TD Nolin, VL Ellingrod.
Pharmacotherapy: A Pathophysiologic Approach. 11th Edition.
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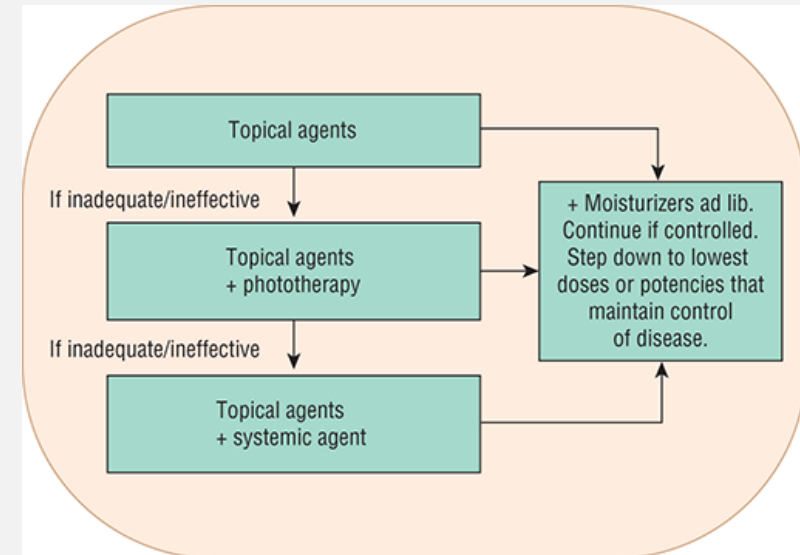


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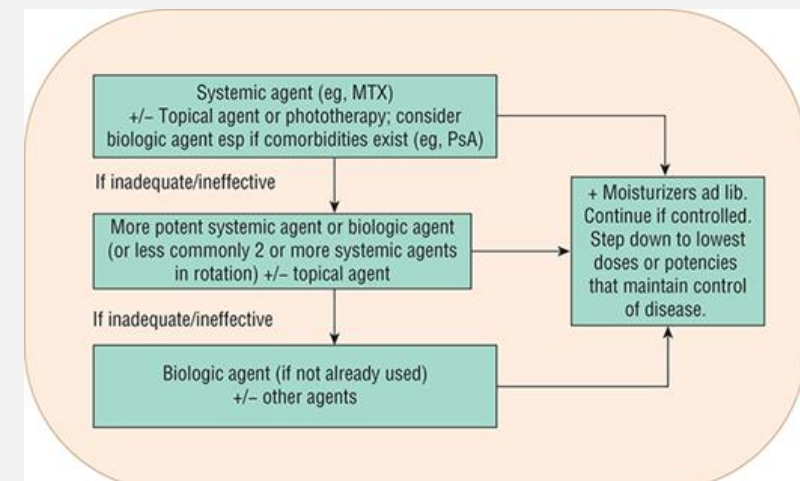
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Pharmacotherapy: A Pathophysiologic Approach. 11th Edition.
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TREATMENT OF PSORIASIS

- topical agents (approach of choice for mild-to-moderate)
 - retinoids e.g., **tazarotene**
 - Vit D analogs **calcipotriene, calcitriol**
 - corticosteroids e.g., **betamethasone**
 - reduce after lesions diminish to avoid CS-induced skin atrophy
- phototherapy (moderate-to-severe)
 - UVB narrow or wide “band”
- photochemotherapy (moderate-to-severe)
 - UVA + **psoralen**
- systemic agents (approach of choice for extensive or moderate-to-severe)
 - methotrexate
 - retinoids, e.g., **acitretin**
 - cyclosporine
 - biologics
 - **note** **apremilast**
 - (PDE4 inhibitor: for mild-to-moderate not controlled by topical agents (UptoDate Jan 2022))



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mild



severe

TREATMENT OF ATOPIC DERMATITIS/ECZEMA

ADULT

(a) Treatment recommendation for atopic eczema: adult

- For every phase, *additional* therapeutic options should be considered
- Add antiseptics / antibiotics in cases of superinfection
- Consider compliance and diagnosis, if therapy has insufficient effect
- Refer to guideline text for restrictions, especially for treatment marked with¹
- Licensed indication are marked with², off-label treatment options are marked with³

SEVERE:
SCORAD >50 / or
persistent eczema

Hospitalization; systemic immunosuppression:
cyclosporine A², short course of oral
glucocorticosteroids², dupilumab^{1,2}, methotrexate³,
azathioprin³, mycophenolate mofetil³, PUVA¹,
alitretinoin^{1,3}

MODERATE:
SCORAD 25-50 / or
recurrent eczema

Proactive therapy with topical tacrolimus² or class
II or class III topical glucocorticosteroids³, wet wrap
therapy, UV therapy (UVB 311 nm, medium dose UVA1),
psychosomatic counseling, climate therapy

MILD:
SCORAD <25 / or
transient eczema

Reactive therapy with topical glucocorticosteroids class
II² or depending on local cofactors: topical calcineurin
inhibitors², antiseptics incl. silver², silver coated textiles¹

corticosteroids

BASELINE:
Basic therapy

Educational programmes, emollients, bath oils,
avoidance of clinically relevant allergens (encasings, if
diagnosed by allergy tests)

lifestyle changes

PEDIATRIC

(b) Treatment recommendation for atopic eczema: children

- For every phase, *additional* therapeutic options should be considered
- Add antiseptics / antibiotics in cases of superinfection
- Consider compliance and diagnosis, if therapy has insufficient effect
- Refer to guideline text for restrictions, especially for treatment marked with¹
- Licensed indication are marked with², off-label treatment options are marked with³

SEVERE:
SCORAD >50 / or
persistent eczema

Hospitalization, systemic immunosuppression:
cyclosporine A³, methotrexate³, azathioprin³,
mycophenolate mofetil^{1,3}

MODERATE:
SCORAD 25-50 / or
recurrent eczema

Proactive therapy with topical tacrolimus² or class II or
III topical glucocorticosteroids³, wet wrap therapy, UV
therapy (UVB 311 nm)¹, psychosomatic counseling,
climate therapy

MILD:
SCORAD <25 / or
transient eczema

Reactive therapy with topical glucocorticosteroids class
II² or depending on local cofactors: topical calcineurin
inhibitors², antiseptics incl. silver, silver coated textiles

BASELINE:
Basic therapy

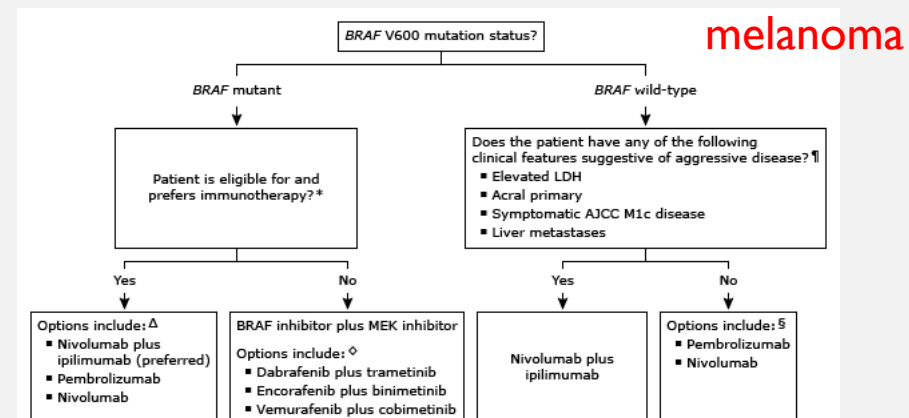
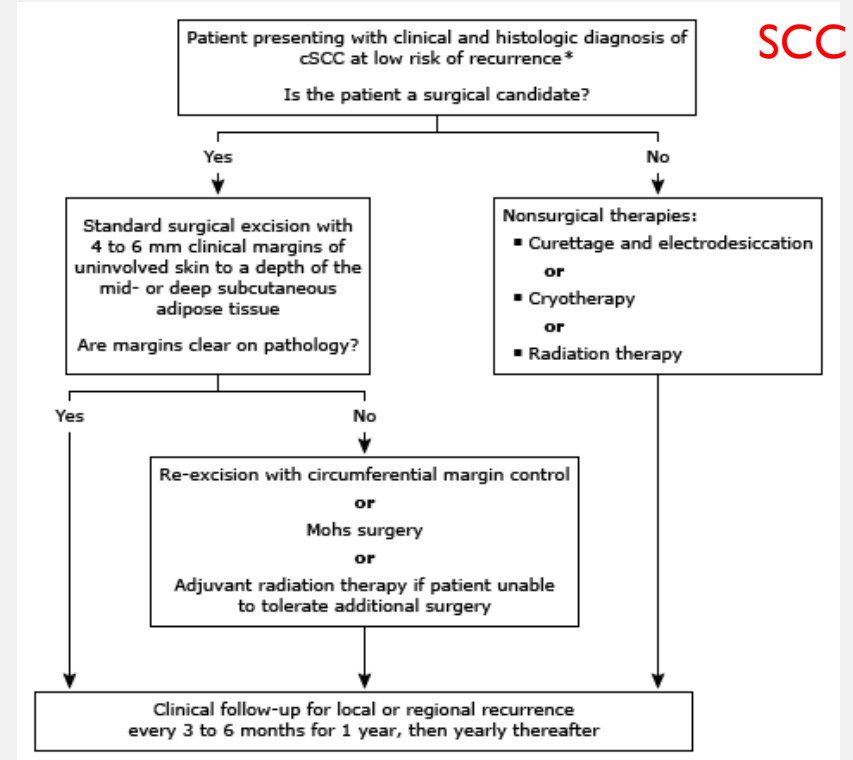
Educational programmes, emollients, bath oils, avoid-
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note tacrolimus, dupilumab, retinoids, methotrexate

TREATMENT OF DERM NEOPLASMS

- **systemic retinoids**
 - acitrecin, isotretinoin, alitretinoin, bexarotene
- topical **5-FU**
 - diffuse SCC, BCC
- **Hedgehog pathway inhibitors** (HH active in **BCC**)
 - vismodegib, sonidegib
- imiquimod
 - Toll-like receptor 7 agonist
 - promotes inflammatory cell infiltration and apoptosis
- other immunomodulators (see flowcharts →)



HYPERPIGMENTATION CORRECTION

- post-inflammatory hyperpigmentation (PIH)
 - acne, eczematous dermatoses, injury, psoriasis, atopic dermatitis
- melasma
 - photodamage, inflammation, hormone and genetic contributors
- general treatment goal: **inhibition of melanin synthesis/destruction of melanocytes**
- **Treatment approaches:**
 - photoprotection (**preventive**)
 - broad-spectrum sunscreen preferred
 - containing **iron oxide** (improves UVA-blocking)
 - **hydroquinone, monobenzone**
 - inhibits tyrosinase (melanin synthesis)
 - monobenzone may also destroy melanocytes
 - **Vit. C, retinoids**
 - **azelaic acid** (inhibits tyrosinase)
 - **thiamidol** (inhibits tyrosinase)

- **NOT an approved medication: mercury: BANNED WORLDWIDE, NEVER INDICATED, HIGHLY TOXIC!!!!**
 - but may still appear in some cosmetic/unregulated formulations despite being widely banned (WHO)
 - sneaky keywords “antiseptic”, “brightening”, “beautifying”, “purifying”, “balancing”
- see recent public health press releases from [MN](#), [TX](#)
- “People should not buy or use products that may have the following words in ingredient list- “mercury,” “mercurio,” “calomel,” or mercury compounds such as “mercurous chloride.””



PIH

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UptoDate



melasma

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UptoDate

TX Dept of HHS

- hydroquinone:
 - irritation, abnormal **repigmentation** (esp in sun-exposed skin), exogenous **ochronosis**, potential carcinogen (but not established in humans)
- other skin lightening formulations:
 - corticosteroids: **skin atrophy**, striae, HPA suppression
 - mercury, arsenic: **highly toxic**

QUICK REVIEW

- What aspect(s) of acne occurrence do retinoids help to treat?
- What drug class would you most likely use to treat:
 - candidiasis?
 - ringworm?
 - head lice?
 - herpes sore?
- A patient presents to the ER with a high fever and extensive blistering on the face, neck, chest, and arms, about 7 days after starting a course of SMX-TMP to treat a lingering URTI. What condition do you suspect and is it serious?

QUICK REVIEW ANSWERS

- What aspect(s) of acne occurrence do retinoids help to treat?
 - bacterial proliferation, inflammation, abnormal keratinization/desquamation
- What drug class would you most likely use to treat:
 - candidiasis?
 - antifungal
 - ringworm?
 - antifungal
 - head lice?
 - pediculicide
 - herpes sore?
 - antiviral
- A patient presents to the ER with a high fever and extensive blistering on the face, neck, chest, and arms, about 7 days after starting a course of SMX-TMP to treat a lingering URTI. What condition do you suspect and is it serious?
 - note fever, plus symptoms consistent with Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN); YES IT IS VERY SERIOUS

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

~~———— PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common conditions~~

~~———— acne~~

~~———— psoriasis~~

~~———— eczema/atopic dermatitis~~

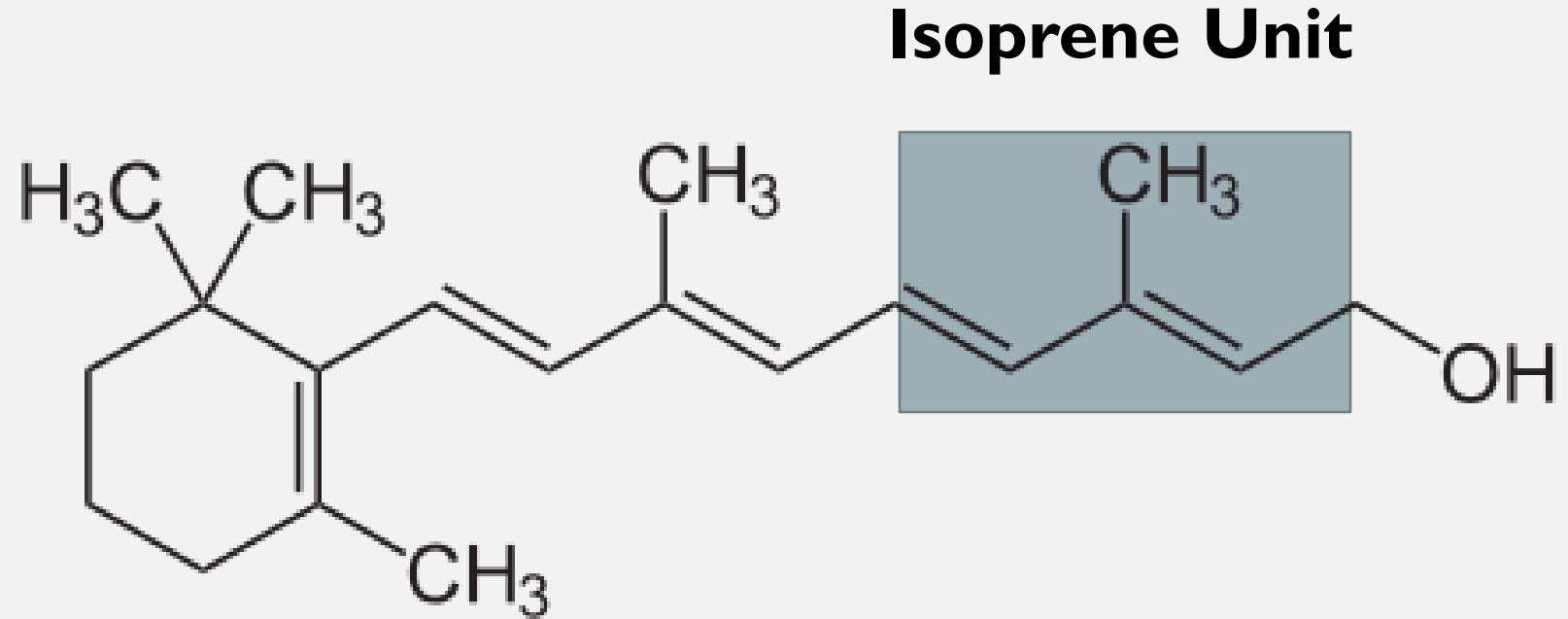
~~———— neoplasms~~



Retinoids

Photochemotherapy

Photoprotection: sunscreens



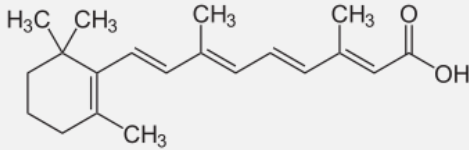
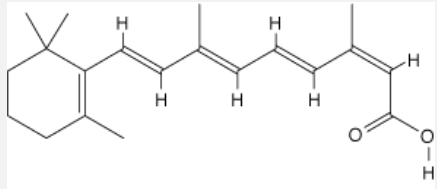
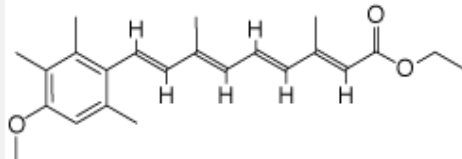
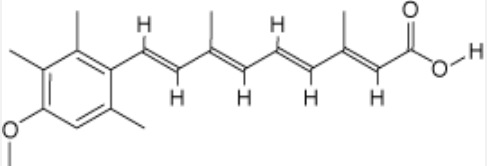
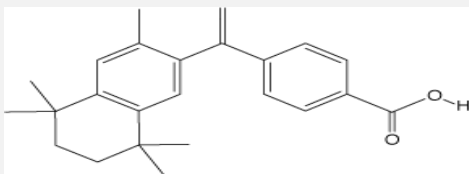
Beta-ionone ring

Isoprenoid: Multiple Isoprene Units

Vitamin A: retinol

TYPES OF RETINOIDS

- **1st generation retinoids** include **retinol** (vitamin A), **tretinoin** (all-*trans*-retinoic acid; vitamin A acid), **ISOTRETINOIN** (13-*cis*-retinoic acid), and **ALITRETINOIN** (9-*cis*-retinoic acid).
- **2nd generation retinoids** (aka *aromatic retinoids*) include **acitretin** and **etretinate**.
 - note very long half lives
- **3rd generation retinoids** include **tazarotene**, **bexarotene**, and **adapalene**.
 - note optimized for receptor-selective binding

retinoids	STRUCTURE	t _{1/2}
retinoic acid		1-2 hr
Isotretinoin		10-20 hr
Etretinate		120 days
Acitretin		50 hours
Bexarotene		7-9 hours

note very long t_{1/2}

- **Receptor Activation**

- **RAR- Retinoic Acid Receptor**

- Binding by retinoids affects cell proliferation signals*

- **RXR- Retinoid X Receptor**

- Binding by retinoids activates apoptosis*

- **VDR- Vitamin D Receptor**

- like RAR, VDR forms a heterodimer with RXR; complex tx outcomes

- **RAR and RXR both have α , β , γ forms**

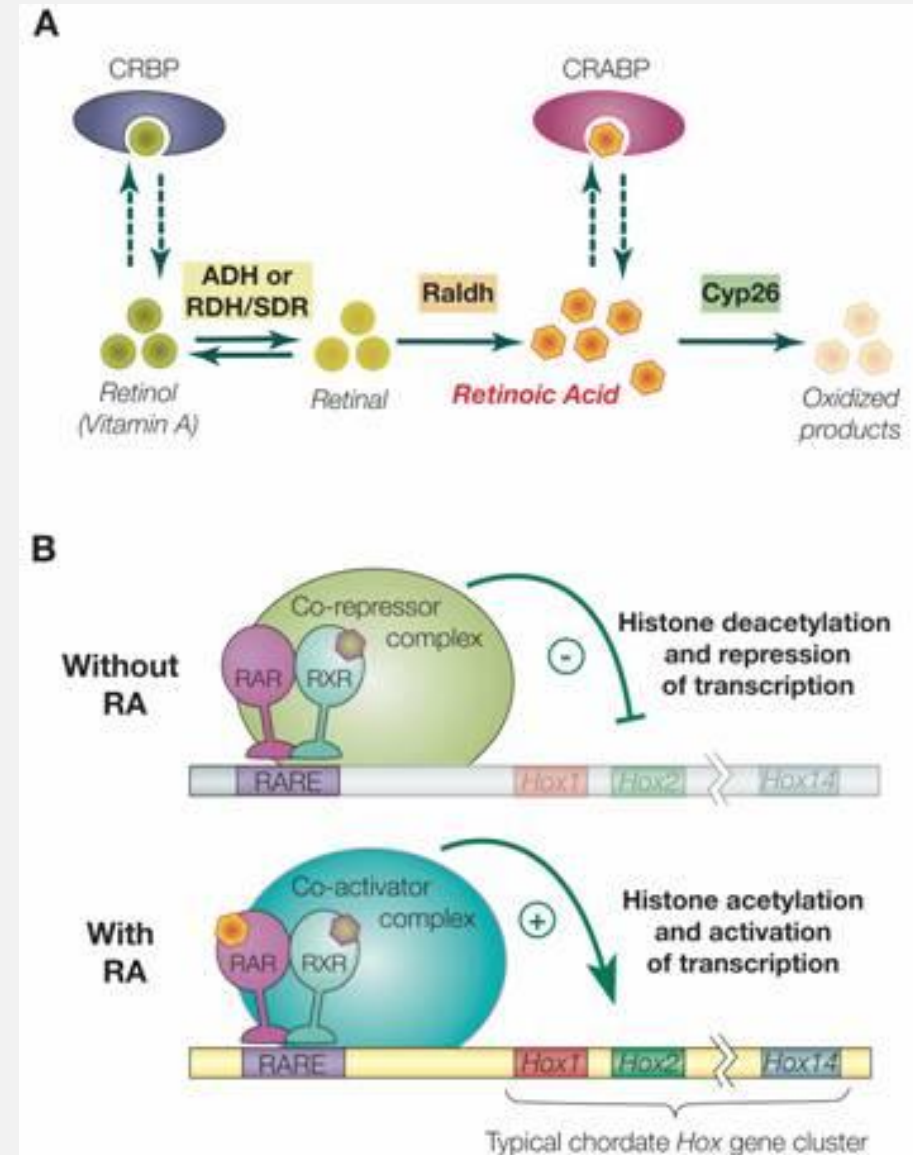
- **RAR**

- drives cell differentiation/proliferation*
 - **Tretinoin, isotretinoin, adapalene, tazarotene**

- **RXR**

- drives apoptosis*
 - **Bexarotene, alitretinoin**

* IN GENERAL, not in all cases



- **Photoaging**

- UV irradiation of human skin damages the dermal matrix in a complex manner that is not completely understood.
- However, topical **tretinoin** partially reverses **collagen deficiency, improves vascularization, and decreases further matrix degradation** induced by solar UV

- **Acne**

- Retinoids can improve **acne scars** and reduce **postinflammatory hyperpigmentation (PIH)**.

- **Psoriasis**

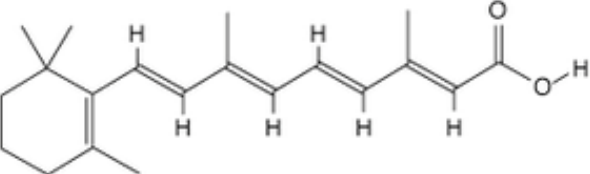
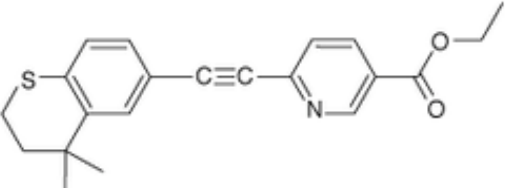
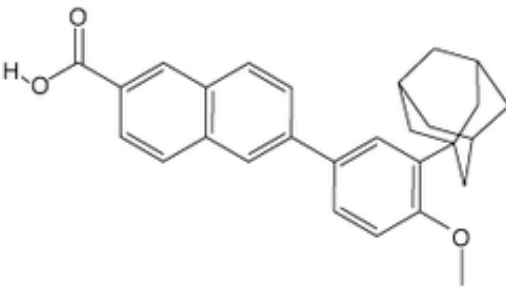
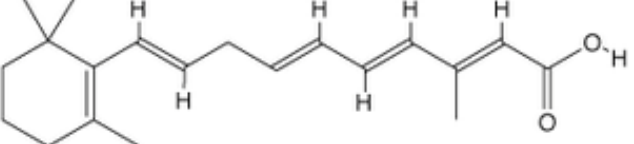
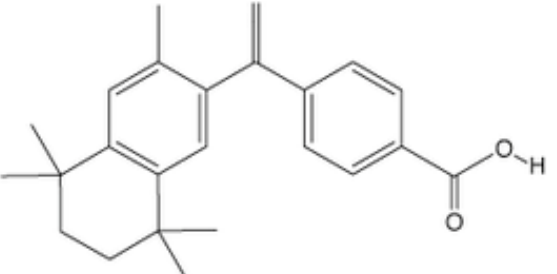
- Retinoids are known to modulate **epidermal proliferation** and **differentiation** and to have **immunomodulatory** and **anti-inflammatory activity**.
- **Soriatane (Acitretin)** is the only oral retinoid approved by the FDA specifically for treating **psoriasis**. The exact way it works to control psoriasis is unknown.

- **Dermatological Cancers**

- The oral retinoid **isotretinoin** (Roaccutane) improves wrinkles and other sun-induced skin damage, while **actinic keratoses** and **basal cell carcinomas** have been treated with the topical retinoid **tretinoin**.

TOPICAL RETINOIDS

29

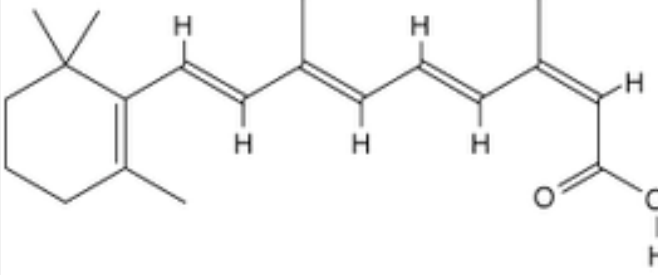
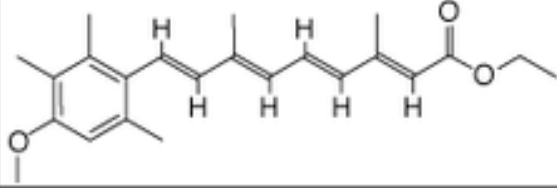
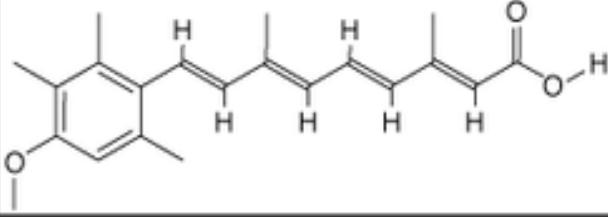
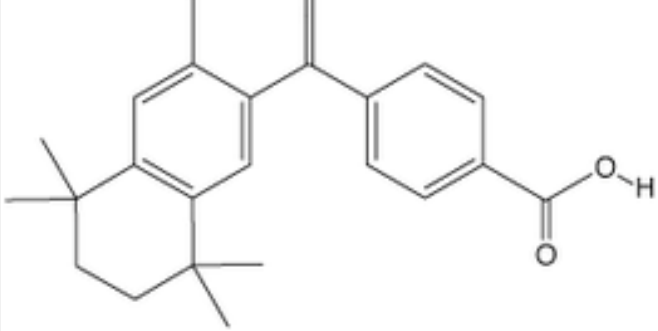
DRUG	FORMULATION	RECEPTOR SPECIFICITY	PREGNANCY CATEGORY	STRUCTURE	IRRITANCY
Tretinoin	0.02%, 0.025%, 0.05%, 0.1% cream .01%, 0.025%, 0.04% gel 0 0.05%, 0.1% solution	RAR- α , RAR- β , RAR- γ	C		++
Tazarotene	0.05%, 0.1% cream 0.05%, 0.01% gel	RAR- α , RAR- β , RAR- γ	X		++++
Adapalene	0.1%, 0.3% cream 0.1%, 0.3% gel 0.1% solution	RAR- β , RAR- γ	C		+
Alitretinoin	0.1% gel	RAR- α , RAR- β , RAR- γ	D		++
Bexarotene	1% gel	RXR- α , RXR- β , RXR- γ	X		+++

note:
irritancy
affects pt
compliance

note: **not** all
topical
retinoids
category X

SYSTEMIC RETINOIDS

30

DRUG	STRUCTURE	RECEPTOR SPECIFICITY	STANDARD DOSING RANGE	t _{1/2}
Isotretinoin		No clear receptor affinity	0.5-2 mg/kg/day	10-20 hours
Etretinate		RAR- α , RAR- β , RAR- γ	0.25-1 mg/kg/day	80-160 days
Acitretin		RAR- α , RAR- β , RAR- γ	0.5-1 mg/kg/day	50 hours
Bexarotene		RXR- α , RXR- β , RXR- γ	300 mg/m ² /day	7-9 hours

note very
long half
lives

- **Topical**
 - UV sensitivity; sunburn
 - Desquamation (peeling)
 - Irritation: dry-feeling skin, burning/stinging
- **Systemic**
 - **Retinoid toxicity**
 - Dry skin, nosebleeds
 - Hair loss
 - IBS trigger
 - Muscle and joint pain (myalgia, arthralgia)
 - **Potent teratogen:** pregnancy category X
 - RAR/RXR control many aspects of embryonic development
 - **Hypothyroidism**
 - RXR-dependent *TSH* gene suppression

RETINOIDS ADVERSE EFFECTS

- Oral (systemic) retinoids are known teratogens: **absolute C/I during pregnancy**
 - effective contraception MUST be used concurrently with systemic retinoids.
- Ok, but what about topical retinoids?
 - some evidence that topical retinoids may also be teratogenic
 - E.g., topical [tretinoin](#) (Stieva-A) is “not recommended during pregnancy or in women of childbearing years without the proper use of an effective method of contraception.” mfr label
 - Nonclinical reproductive toxicity studies have detected developmental toxicity at doses ≥ 80 -fold of the anticipated topical [tretinoin](#) clinical dose. Although systemic absorption from topical [tretinoin](#) is between 1% to 6% when properly used, risk cannot be excluded

note: **not** all topical
retinoids category X

note: **all systemic** retinoids
are pregnancy category X

BOARDS-STYLE EXAMPLE

A 13-year-old female patient presents to her primary care physician for a follow-up of her treatment for acne vulgaris. The patient is currently using benzoyl peroxide, topical retinoids, and erythromycin. Unfortunately, she does not seem to be responding to the current treatment. The patient is placed on treatment with oral retinoids. Prior to initiating therapy, she and her parents are warned of the common side effects of this treatment, which include:

- A. dry skin and muscle pains
- B. encephalitis
- C. increased incidence of cholangiocarcinoma
- D. increased incidence of hepatocellular carcinoma
- E. pancreatitis

Bottom Line: Oral isotretinoin produces predictable and manageable side effects that are, for the most part, reversible on discontinuation of therapy. Most are similar to those seen in high dose vitamin A therapy that includes dry cracked lips, xerosis of the skin, mucous membranes and eyes, myalgia and arthralgia, which tend to be transient and dose-related to exercise, and elevated levels of lipids and liver enzymes have been associated with therapy.

QUICK REVIEW

- How do 3rd-gen retinoids tend to differ from previous generations?
Give one example of a 3rd-gen retinoid.
- What is the two-word description of retinoids MOA?
- A primarily RAR-interacting retinoid probably drives which of the following, apoptosis or differentiation?
- What is an absolute contraindication for oral etretinate?

QUICK REVIEW

- How do 3rd-gen retinoids tend to differ from previous generations?
Give one example of a 3rd-gen retinoid.
 - greater receptor selectivity; e.g., tazarotene, bexarotene, adapalene
- What is the two-word description of retinoids MOA?
 - transcriptional regulation
- A primarily RAR-interacting retinoid probably drives which of the following, apoptosis or differentiation?
 - differentiation
- What is an absolute contraindication for oral etretinate?
 - pregnancy or plans to become pregnant

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

~~———— PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common conditions~~

~~———— acne~~

~~———— psoriasis~~

~~———— eczema/atopic dermatitis~~

~~———— neoplasms~~

~~Retinoids~~

 **Photochemotherapy**

Photoprotection: sunscreens

FYI: WHEN NATURE IS NOT YOUR FRIEND

related
furanocoumarin-
containing plants:

- carrot
- parsnip
- Queen Anne's lace
- celery
- dill
- fennel
- parsley

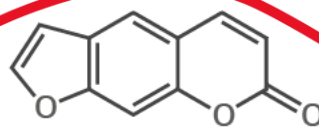
HOW DOES GIANT HOGWEED CAUSE SKIN BURNS?



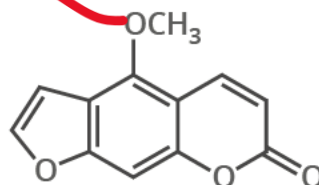
GIANT HOGWEED
HERACLEUM MANTEGAZZIANUM
HEIGHT: UP TO 5.5 METRES

Giant Hogweed is a plant that is originally native to central Asia. In the 19th century it was introduced to the UK as an ornamental plant. Subsequently, it has spread to parts of the USA, Canada, and Europe.

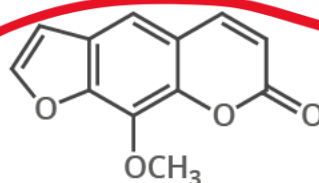
FURANOCOUMARINS



PSORALEN



BERGAPTEN



METHOXSALEN



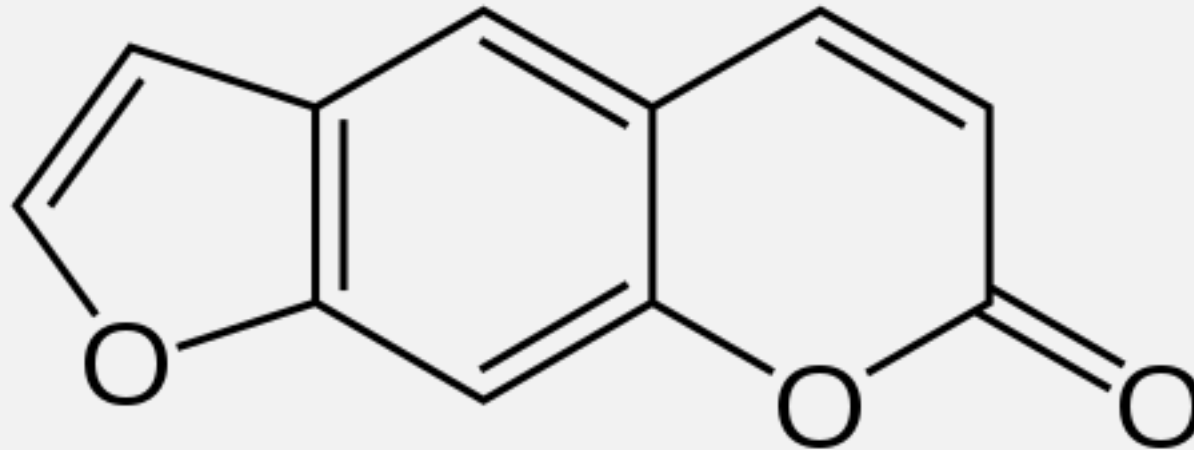
Giant Hogweed's sap contains phototoxic compounds called furanocoumarins (also known as furocoumarins). They are found in all parts of the plant, but the highest levels are found in the leaves. When in contact with the skin, and exposed to UV radiation from sunlight with a wavelength of 320-380 nanometres, they can cause phytophotodermatitis (skin inflammation and burns).



The phototoxic effects of furanocoumarins occur due to their ability to react with bases in DNA to form adducts in the presence of UV-A radiation. These adducts can then react further with other bases to form crosslinks between DNA strands. These crosslinks lead to the characteristic blisters seen on exposure to Giant Hogweed sap.



- **Psoralen** (also called psoralene) is the parent compound in a family of natural products known as **furanocoumarins**.



- It is widely used in P-UVA (psoralen + UVA) treatment for e.g., psoriasis, eczema, vitiligo, and cutaneous T-cell lymphoma.

PHOTO**CHEM**OTHERAPY = UV + PSORALEN

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- UV Light (320-340nm) **in conjunction** with **photosensitizing** agents (**psoralens**)

study q:
which
type of
UV is this?

- Oral delivery followed by UV exposure

- MOA: UV-activated drugs **intercalate** into skin cell DNA, promoting cell alteration and/or **apoptosis (cell death)**.

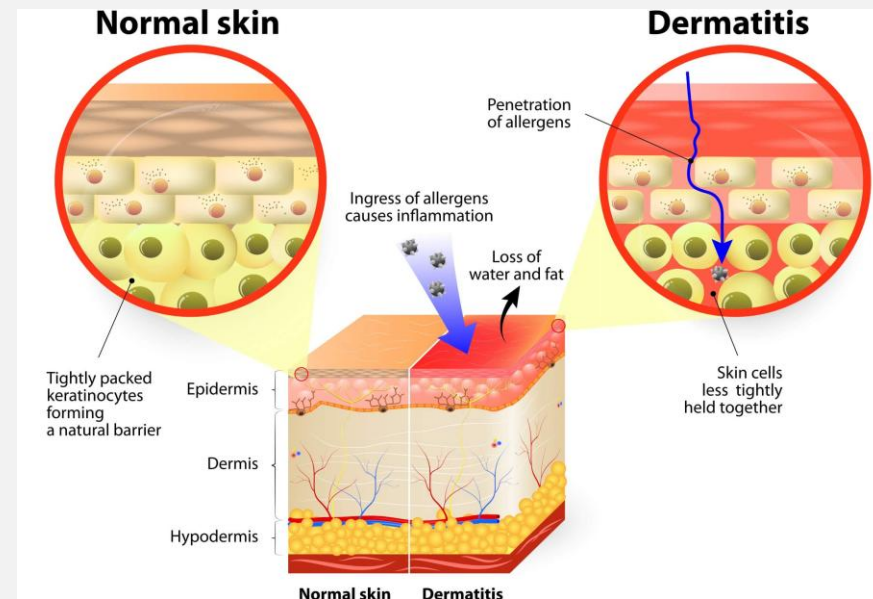
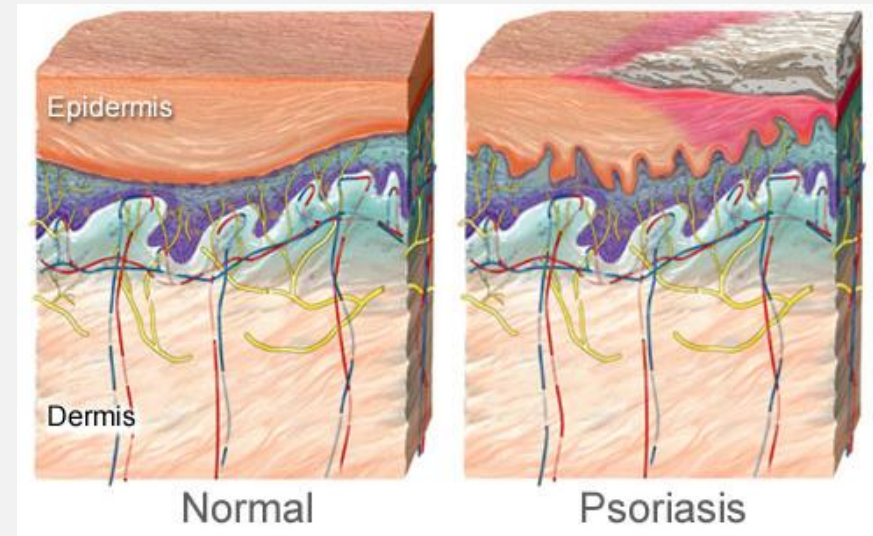
study q:
which
type rxn
is this?

- Also induces ROS production, which is pro-apoptotic.

study q:
which
type rxn
is this?

- **Indications:**

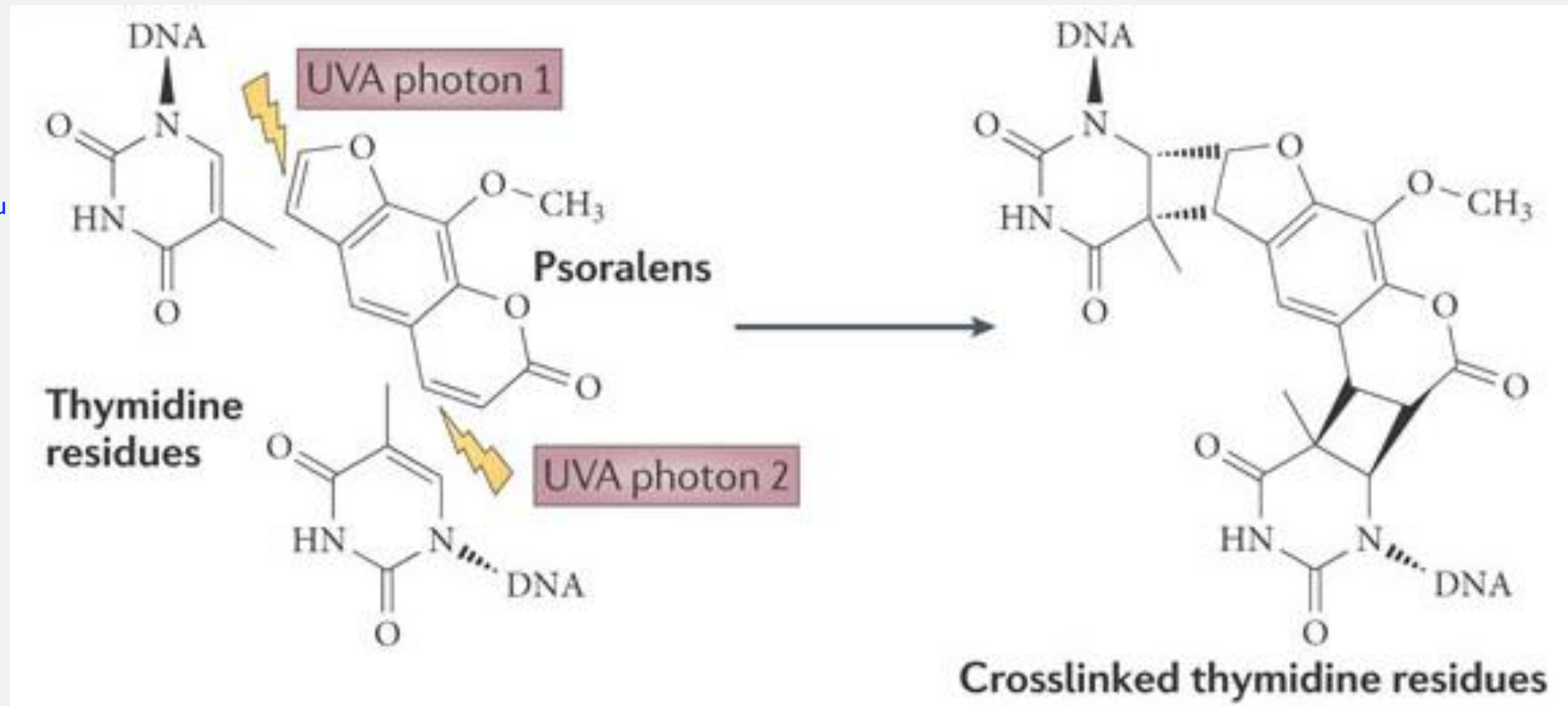
- **Psoriasis**
- **Atopic Dermatitis**
- **Skin Cancers**
- **hypopigmentation correction, vitiligo**
 - vitiligo: autoimmune destruction of melanocytes
 - Non-pigmented patches of otherwise normal skin



Winnie Harlow, Cosmopolitan

PSORALENS MOA **TYPE I** : DNA INTERCALATION

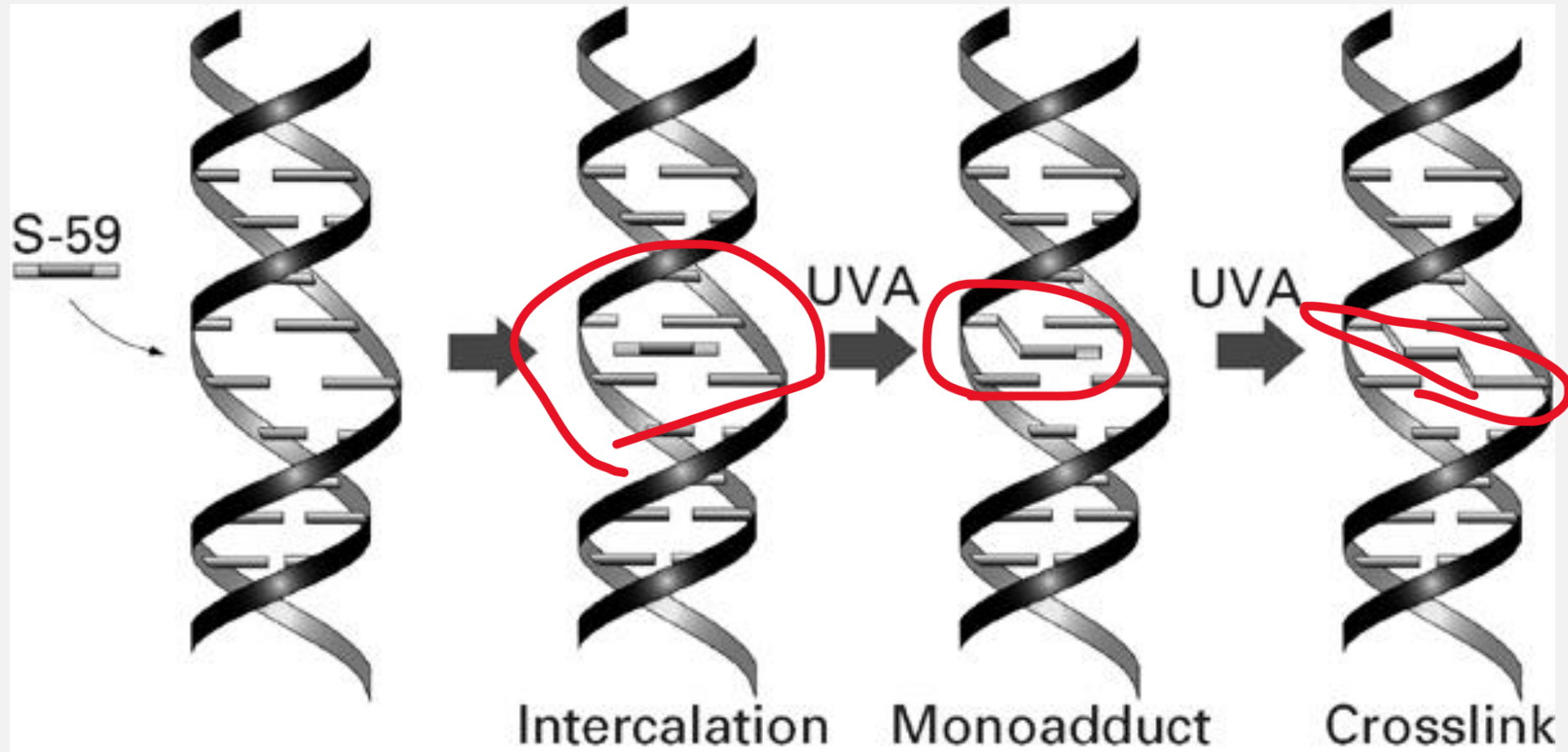
- UV photo-activation reaction with **psoralens**.



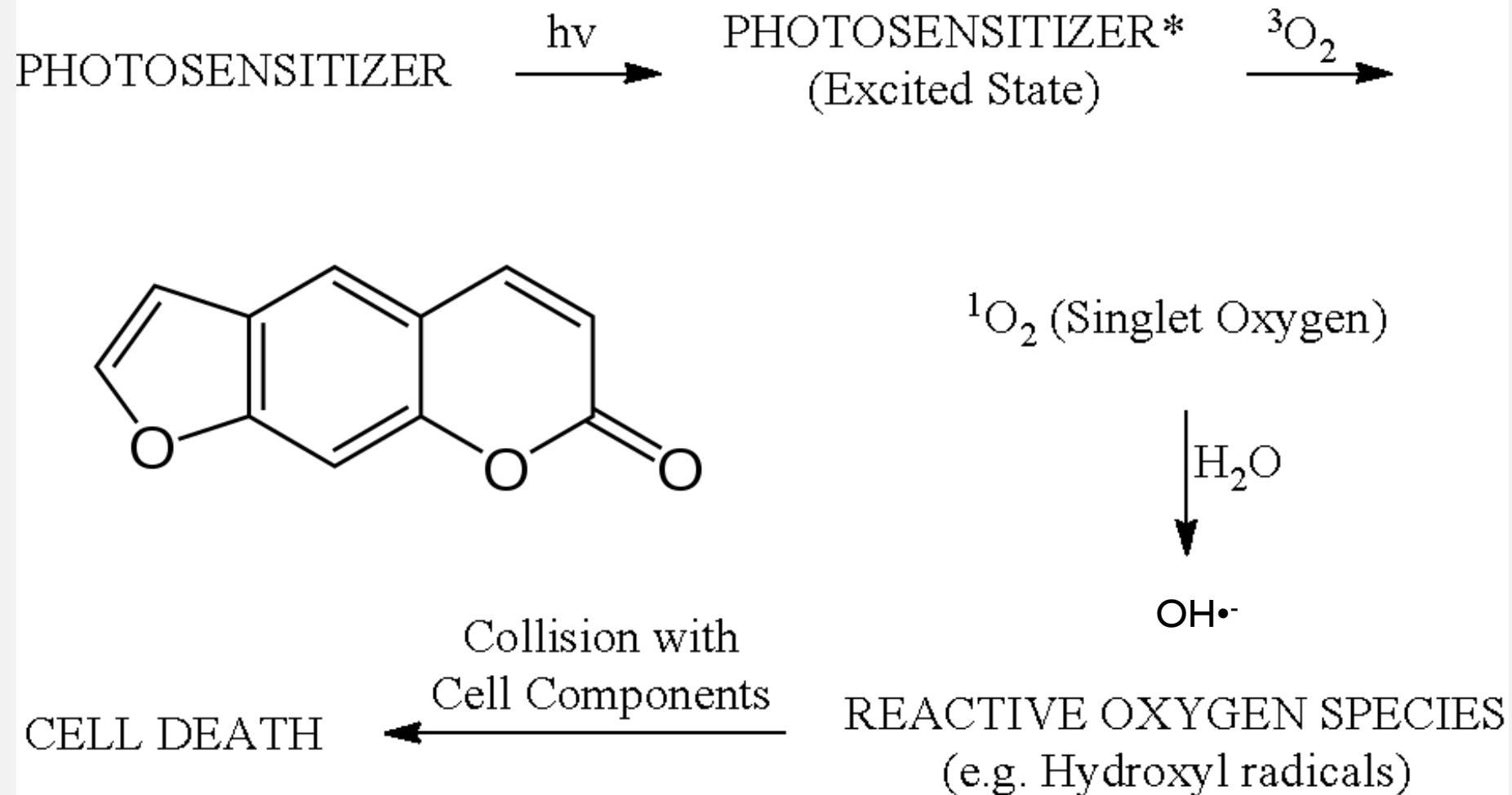
more than you
need to know
study q: which
psoralen is
shown here?
hint- see FYI
slide

PSORALENS TYPE I REACTION: ANOTHER VIEW

41



PSORALENS MOA **TYPE II** : ROS PRODUCTION



- psoralens: amplified photosensitivity in combination with other photosensitizing drugs
 - e.g., phenothiazines, thiazides, sulfonamides, NSAIDs, sulfonylureas, tetracyclines, and benzodiazepines
 - **Phototoxicity**
 - Dry skin, abnormal pigmentation, pruritis, HSV activation
 - Edema, blistering

LECTURE OUTLINE

~~Drug lists (on your own)~~

~~Your skin is a living thing~~

~~———— PK review~~

~~Drug-induced derm emergencies~~

~~Overview of treatments for common co~~

~~———— acne~~

~~———— psoriasis~~

~~———— eczema/atopic dermatitis~~

~~———— neoplasms~~

~~Retinoids~~

~~Photochemotherapy~~

————→ **Photoprotection: sunscreens**

**Can you spot the
dermatologist?**



IF YOUR EYES SAW UV, YOU WOULD SEE THAT SUNSCREEN WORKS

study q: what does this tell you about the mechanism of action of sunscreens?

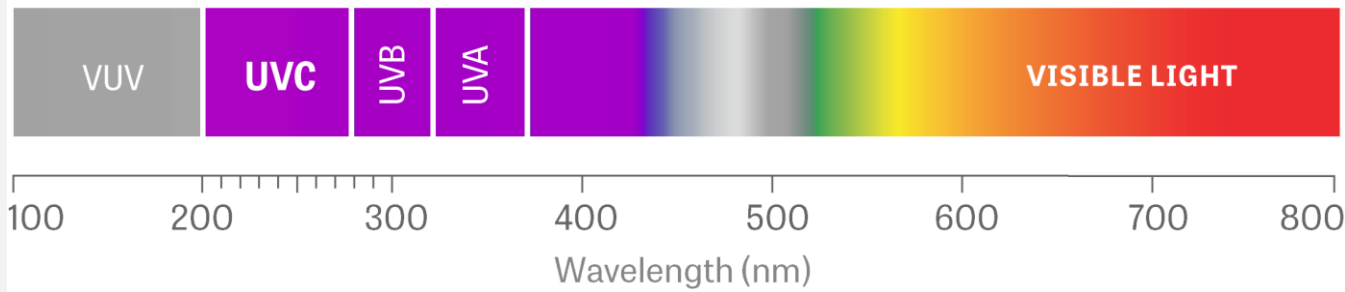


regular camera: detects visible light reflected from surfaces

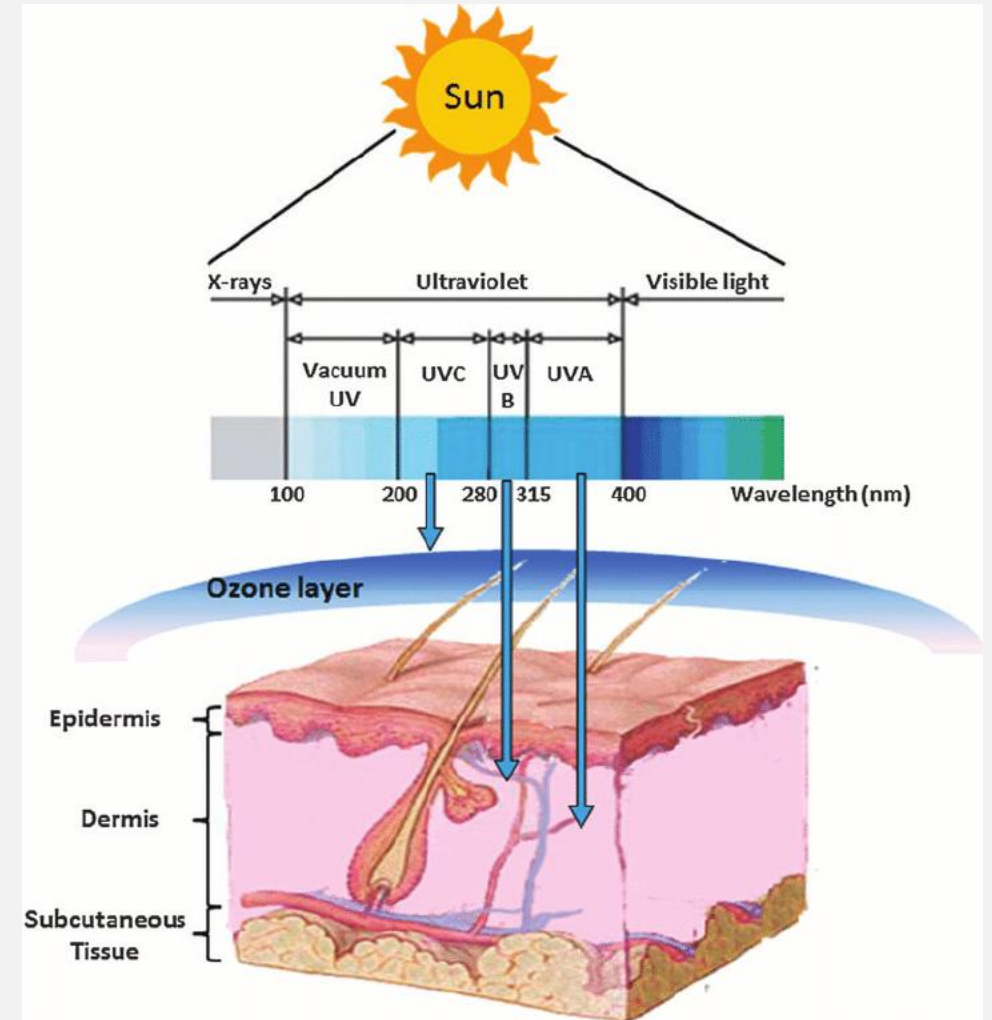
UV camera: detects UV reflected from surfaces

THE UV SPECTRUM: NOT VISIBLE TO HUMANS

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- UV-A: 315-400 nm (**A = Aging**)
 - longest UV nm, deepest skin penetration- contributes to **photoaging, cancer**
 - most UVA reaches Earth's surface
- UV-B: 280-315 nm (**B = Burning**)
 - largest contributor to **sunburn**, also contributes to **cancer**
 - despite ~90% absorbed in atmo
- UV-C: ~200-280 nm (**C = Captured**)
 - ~100% solar UVC absorbed (**captured**) in atmo O₃ layer, BUT
 - used in **laboratory and sterilizing** applications
 - also produced by **arc welders, Hg lamps**



DOI:[10.1089/wound.2012.0366](https://doi.org/10.1089/wound.2012.0366)

MOST CONSUMERS DO NOT UNDERSTAND PHOTOPROTECTION

why? until recently, few products distinguished between UVA and UVB blocking ability; better labeling is now required by the FDA:

Blocking the sun

New federal guidelines for sunscreen labels will give consumers better information about a product's effectiveness.

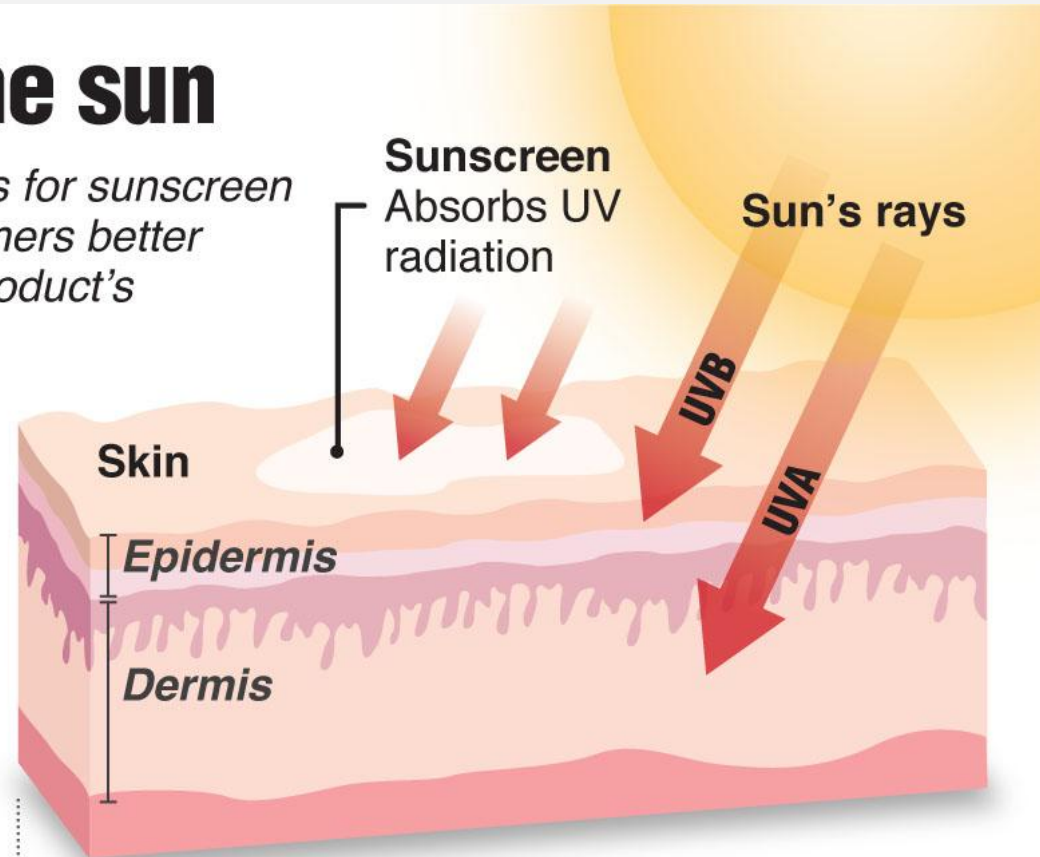
Updated labels

- Products that protect against UVA and UVB will read

Broad spectrum

- Sunscreens that only protect against UVB labels will read

Product has been shown only to help prevent sunburn, not skin cancer or early skin aging



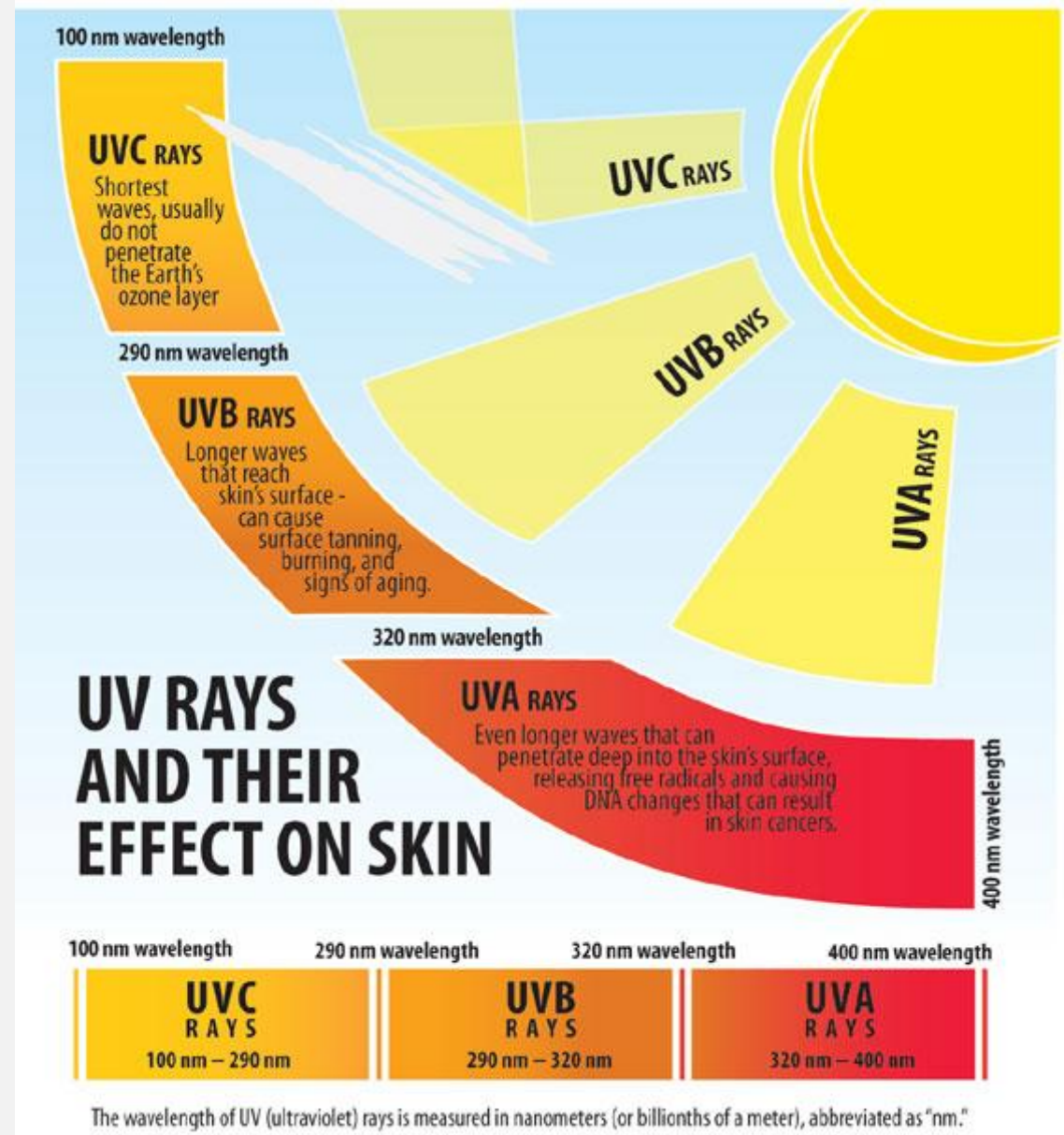
- **UVA rays**
Penetrate deep, weaken tissues; cause cancer

- **UVB rays** Only penetrate epidermis and cause sunburn; SPF only protects against these rays

Source: Skin Cancer Foundation, U.S. Food and Drug Administration
© 2011 MCT
Graphic: Melina Yingling

SUNSCREENS

- The American Cancer Society recommends the use of sunscreen because it aids in the prevention of **squamous cell carcinomas**.
- However, many sunscreens **only block UVB radiation**, and therefore do **not block UVA radiation, which is involved in photoaging and cancers**.
- Specifically, UVA does not primarily cause sunburn but rather can increase the rate of **melanoma** and **photodermatitis**.
- The use of broad-spectrum (UVA/UVB) sunscreens can address this concern.

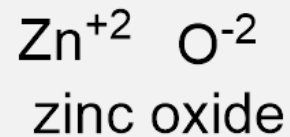


PHOTOBLOCKING INORGANIC FILTERING AGENTS

- photoblocking/inorganic filtering (“Mineral Sunscreens”)

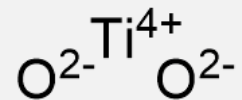
- aka “physical” or “mineral” photoprotectants

- Zinc Oxide



- opaque white

- Titanium Dioxide



titanium dioxide

- opaque white

- **Wide range of wavelengths blocked** (“broad-spectrum”).

- particularly ZnO_2
- note also iron oxides
 - various yellow, brown, red, ochre, black pigments



labmuffin.com



INORGANIC FILTERING

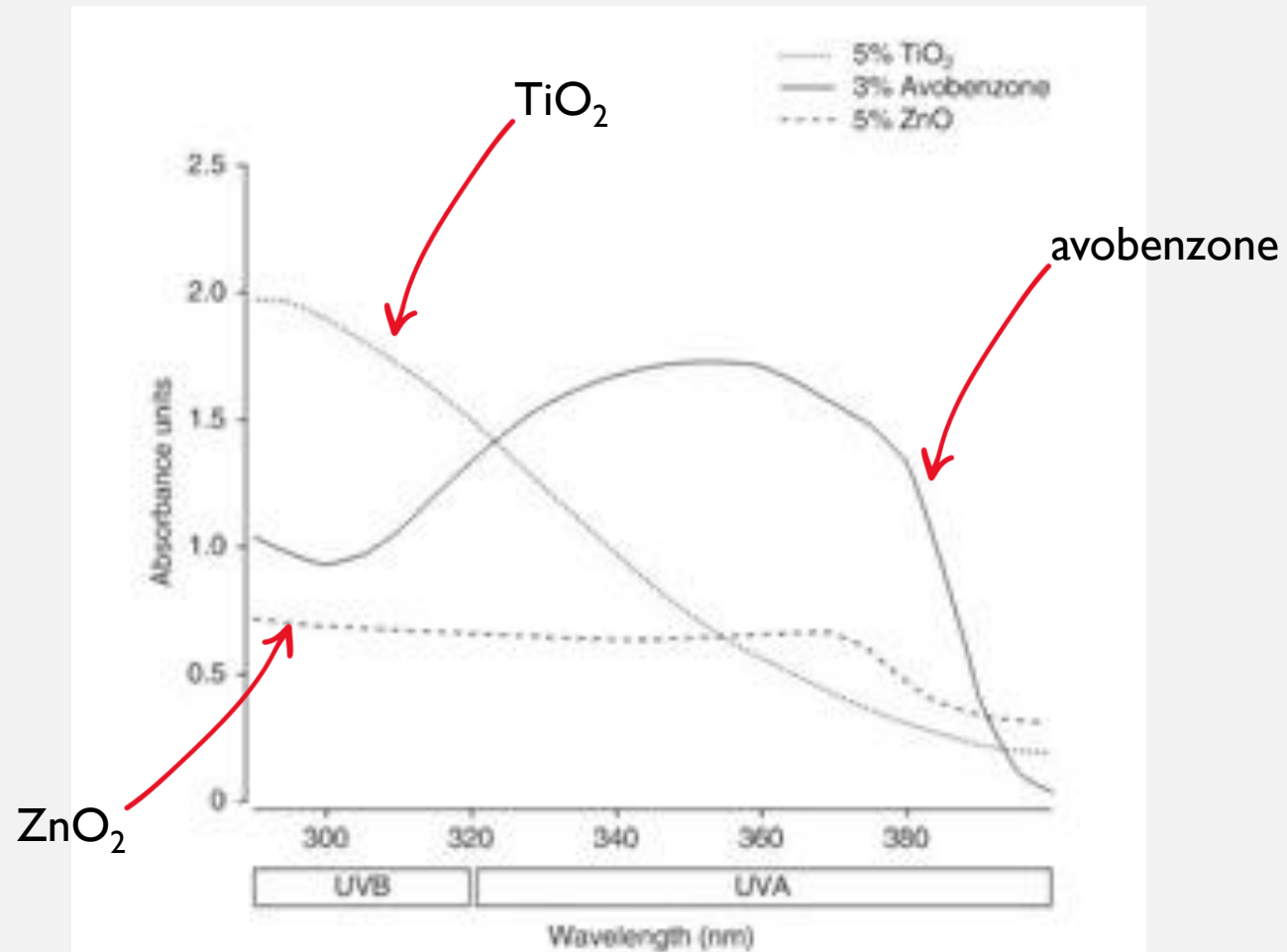
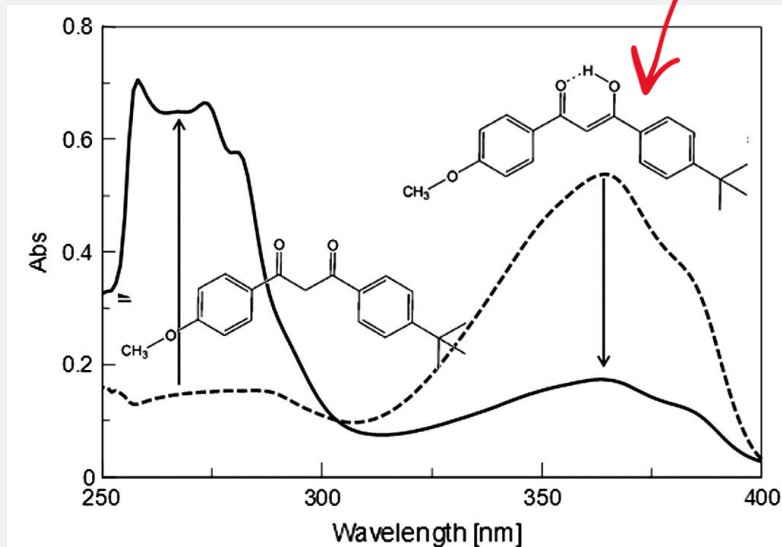
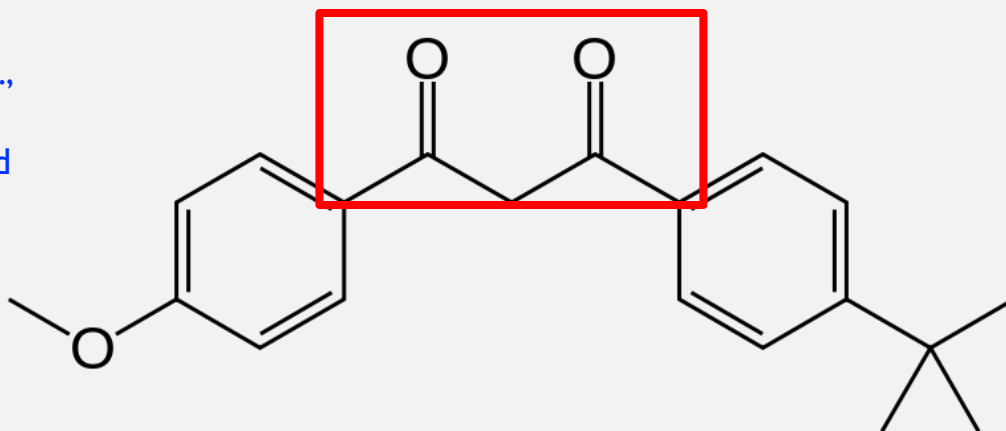


Fig. 1. Comparison of UV attenuation profiles for avobenzone, zinc oxide (ZnO), and titanium dioxide (TiO_2) as single ingredients in a model formulation.

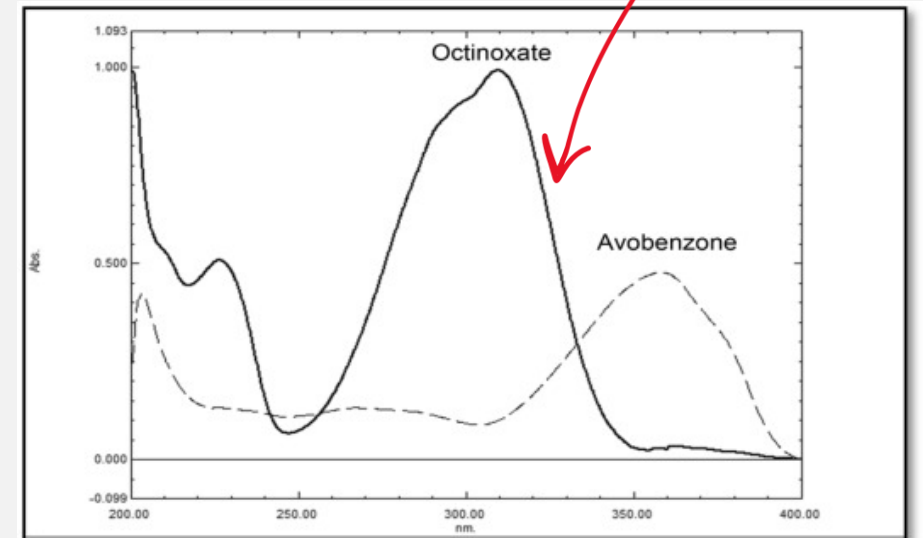
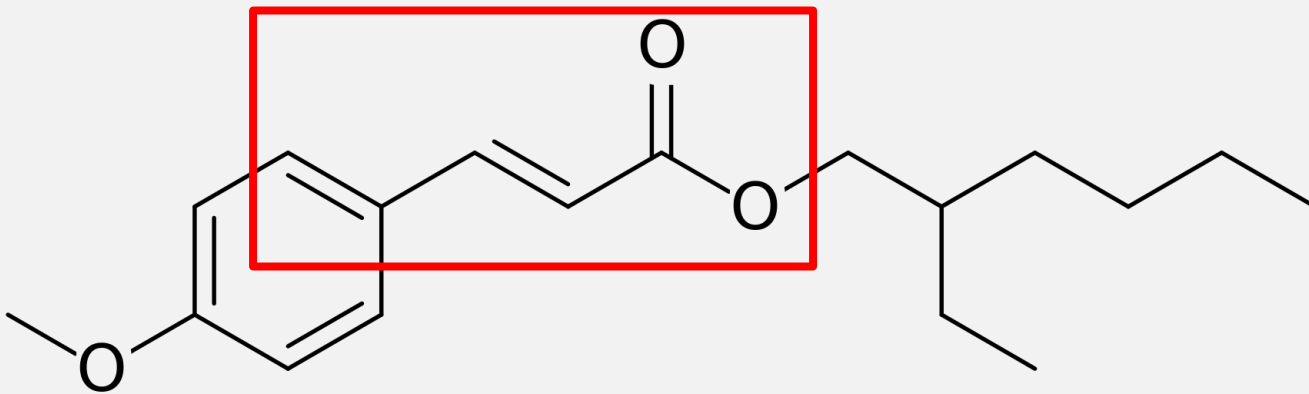
- a.k.a., chemical sunscreens
- **Avobenzene** (1-(4-Methoxyphenyl)-3-(4-*tert*-butylphenyl)propane-1,3-dione) exists in the ground state as a mixture of the enol and keto forms, favoring the chelated enol. This enol form is stabilized by **intramolecular hydrogen-bonding** within the β -diketone. Absorbs ultraviolet light **over a wider range of wavelengths (UVA and UVB)** than many sunscreen agents. Marketed as "broad spectrum" sunscreens. Has absorption maximum of 357 nm.

What's the point here???

This and next 5 slides: the most common chemical sunscreen agents in the U.S.,
Note opportunities for resonance that predict good UV absorbance due to ability to redistribute e^-

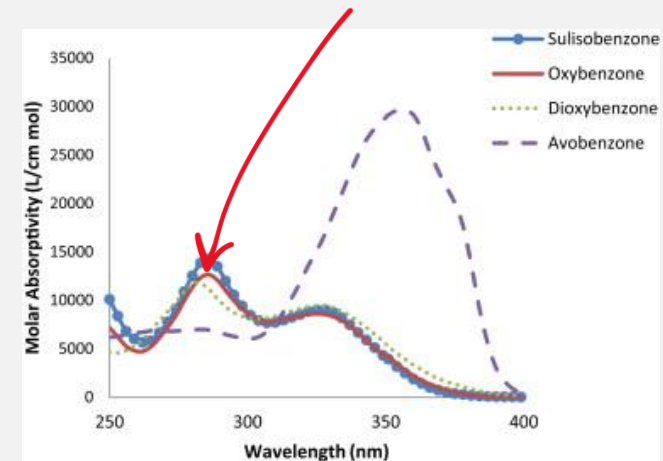
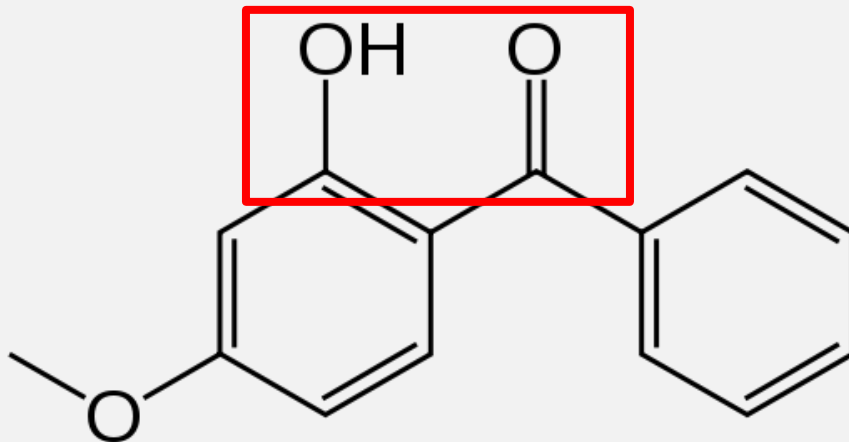


- **Organic Filtering (“Chemical Sunscreens”)**
 - **Octinoxate** (Octyl methoxy**cinnamate** or ethylhexyl methoxy**cinnamate**) is an ester formed from methoxycinnamic acid and (RS)-2-ethylhexanol. Used as an active ingredient in sunscreens combined with **oxybenzone** and **titanium oxide** for its use in protection against UV-B rays.



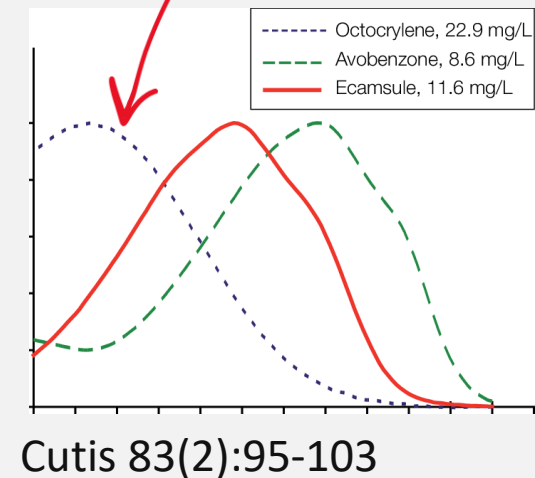
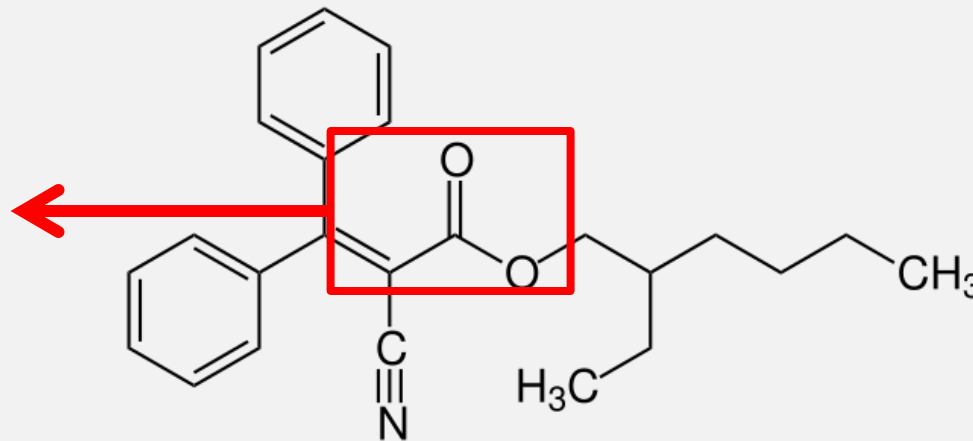
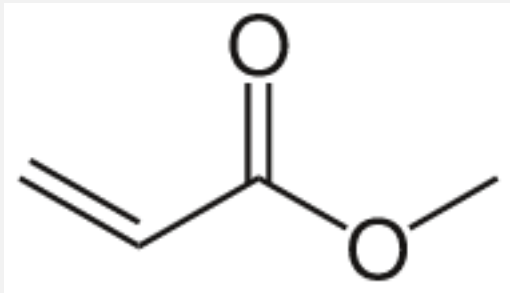
<https://doi.org/10.1016/j.saa.2018.07.073>

- **Organic Filtering (“Chemical Sunscreens”)**
 - **Oxybenzone** (2-Hydroxy-4-methoxyphenyl)-phenylmethanone) provides broad-spectrum ultraviolet coverage, **including UVB and short-wave UVA rays**. It has an absorption profile spanning from 270 to 350 nm with absorption peaks at 288 and 350 nm. Because it is a **conjugated** molecule, oxybenzone absorbs at lower energies than many aromatic molecules. Also, the **hydroxyl group is hydrogen bonded to the ketone**, and this contributes to oxybenzone's light-absorption properties. It is one of the most widely used organic UVA filters in sunscreens today.

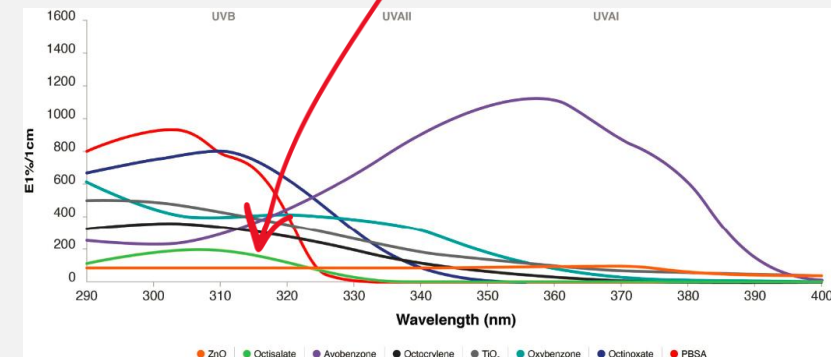
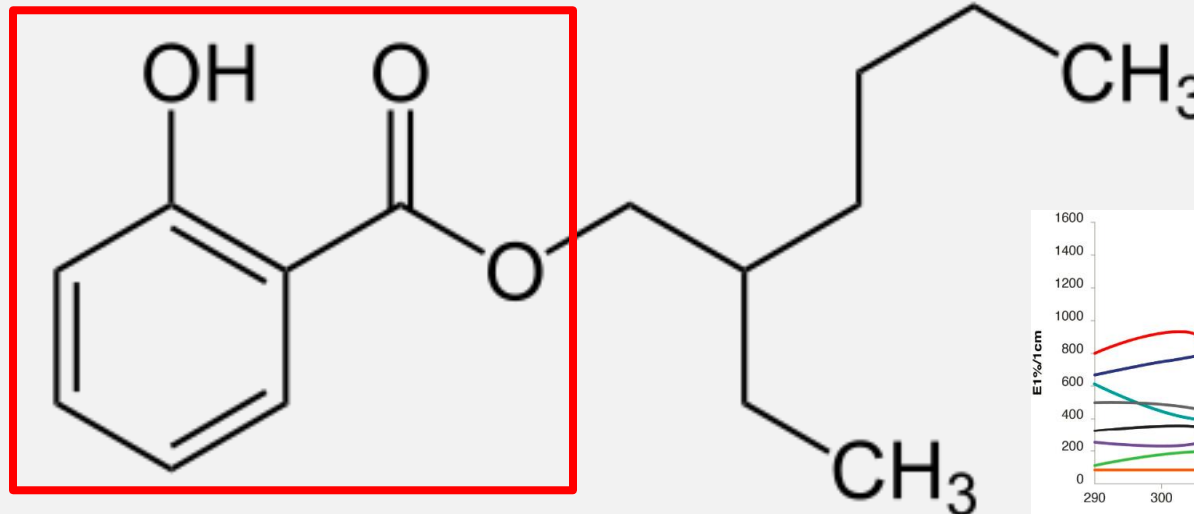
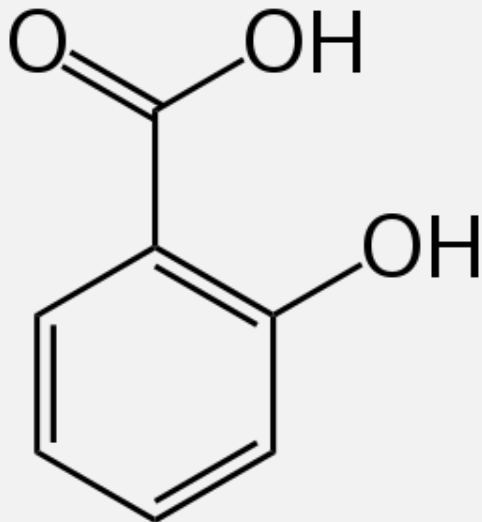


<https://doi.org/10.1016/j.ijpharm.2015.04.027>

- **Organic Filtering (“chemical sunscreens”)**
 - **Octocrylene** is an ester formed by the reaction of 3,3-diphenylcyanoacrylate with 2-ethylhexanol. The **acrylate** portion of the molecule absorbs UVB **and** short-wave UVA (ultraviolet) rays with wavelengths from 280 to 320 nm, protecting the skin from direct DNA damage. **Stabilizes avobenzone** in combined formulation

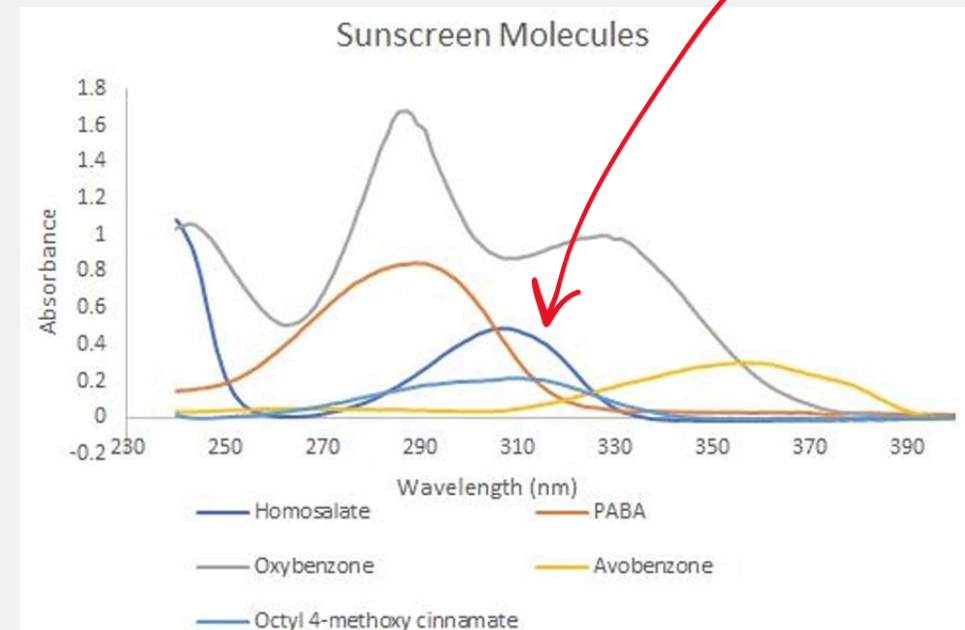
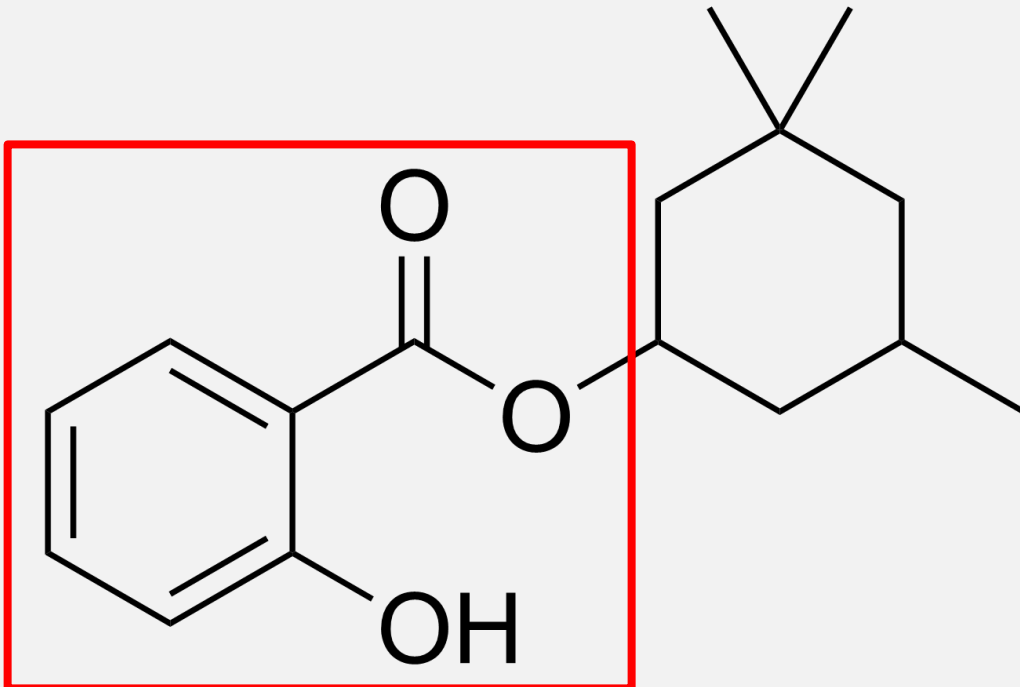


- **Organic Filtering (“Chemical Sunscreens”)**
 - **Octisalate** (Octyl salicylate, or 2-ethylhexyl salicylate), is an ester formed by the condensation of a salicylic acid with 2-ethylhexanol and is used as an ingredient in sunscreens to absorb UVB (ultraviolet) rays. The **salicylate** portion of the molecule absorbs ultraviolet light with a wavelength from 295 nm to 315 nm.

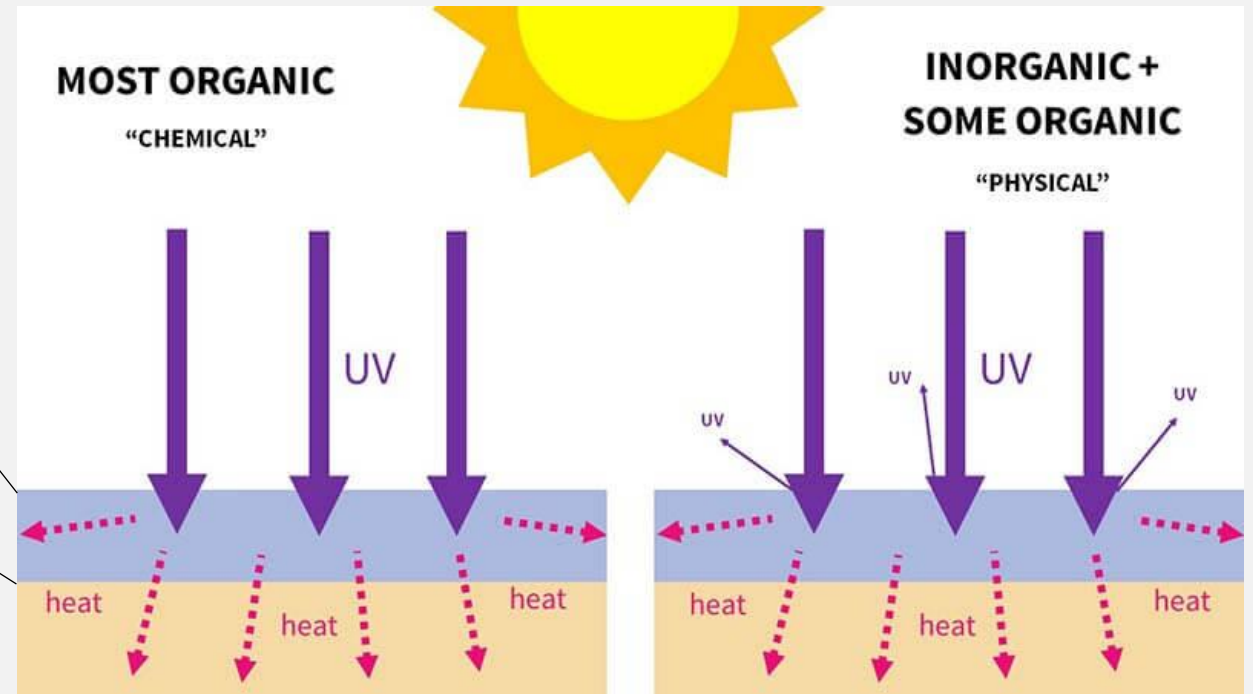
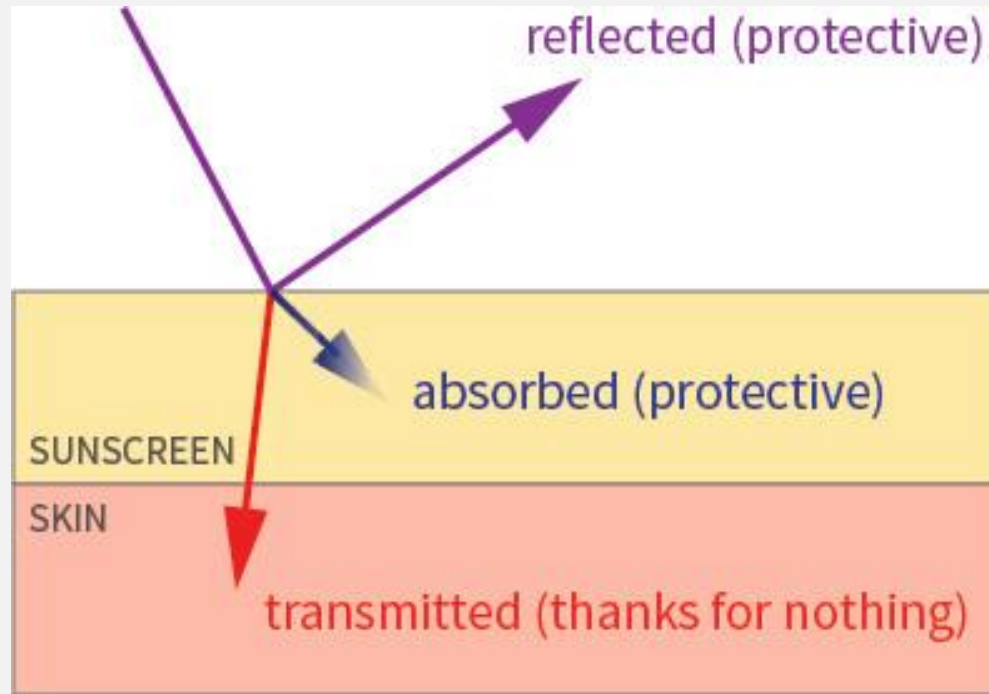


- **Organic Filtering (“Chemical Sunscreens”)**

- **Homosalate** (3,3,5-Trimethylcyclohexyl 2-hydroxybenzoate) is an ester formed from salicylic acid and 3,3,5-trimethylcyclohexanol, a derivative of cyclohexanol. It is **contained in 45% of U.S. sunscreens**. The **salicylic acid** portion of the molecule absorbs ultraviolet rays with a wavelength from 295 nm to 315 nm, protecting the skin from sun damage.



PHYSICAL AND CHEMICAL SUNSCREENS BOTH ABSORB UV RAYS



TiO₂ and ZnO₂ reflect about 4% of UV exposure

Octocrylene	Tinosorb M	A mirror
Tinosorb S		
Octinoxate	Titanium dioxide	
Avobenzone		
Uvinul A Plus	Zinc oxide	
Oxybenzone		
ABSORB MORE		SCATTER & REFLECT MORE

take-home: the MOA of both chemical and physical sunscreens is mostly UV absorption

BEWARE MISINFORMATION



This is the blog of a physician, who should know better.

Grains of truth: organic filters can cross the skin barrier of frozen/thawed pig's ear, a common model system for human skin product testing. separately, rats fed about 1% of their bodyweight in sunscreen over time displayed responses consistent with estrogen signaling.

Misrepresented/Unaddressed: To what extent does this apply to intact human skin in vivo? What are the known biological effects of such absorption in humans applying sunscreen to their skin, not eating 1-2 pounds per day? How does organic filter absorption into the skin compare to mineral absorption?

Take-home: be skeptical of sweeping claims and categorizations. Note lack of context and lack of scale. Note sensationalizing language ("chemicals").

Side note: the same paper cited above states: "Escalation of a phobia towards all organic UV filters is undesirable". The exact opposite of what is happening in this image.

<https://lowchailing.com/2019/05/20/chemicals-in-sunscreen-may-enter-blood/>

SUN PROTECTION FACTOR SIMPLIFIED

SPF-15-30-50 Guide:

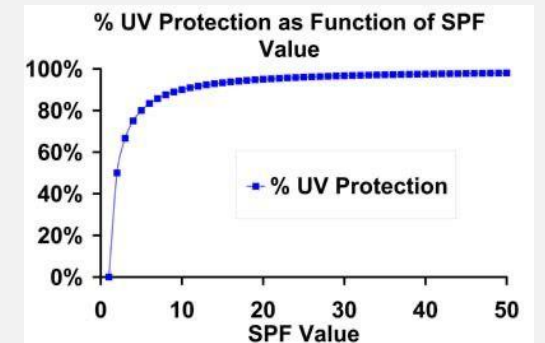
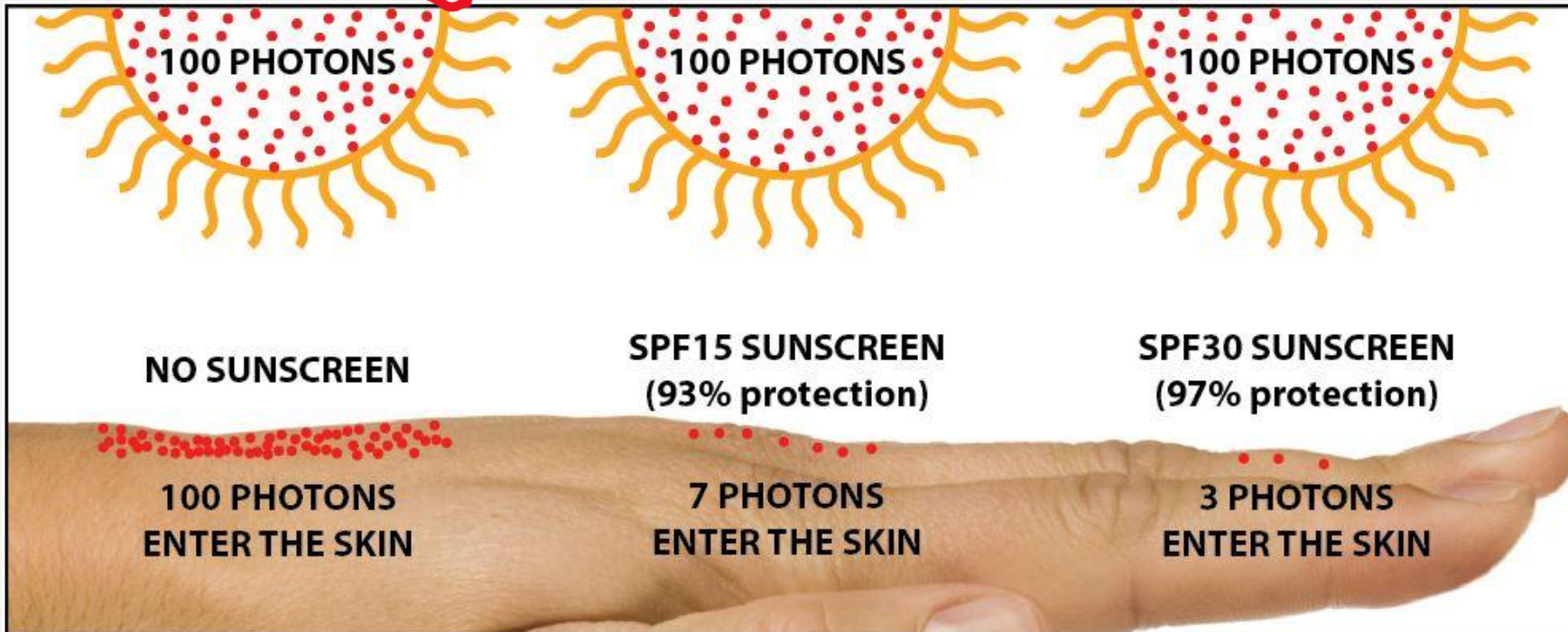
- **15**- 1/15 of the UVB rays get through to your skin - blocking about 93%.
- **30**- 1/30 of the UVB rays get through to your skin - blocking about 97%.
- **50**- 1/50 of the UVB rays get through to your skin - blocking about 98%.

SPF **15** = 93%

SPF **30** = 97%

SPF **50** = 98%

SPF **70** = 98.5%



ACTUAL SPF CALCULATION: IT'S MORE COMPLICATED

- SPF (“Sun Protection Factor”) is a **complex** relationship: oversimplification can lead to inaccuracy.
 - Specifically, the EAS (erythral action spectrum, a measure of tissue damage)
 - also note that testing conditions do not match typical use: much MORE product used in testing than people tend to apply in real life
 - 2-2.5 mL recommended for face and neck; typical use is half or less recommended amt.
- The most important thing to know is that **SPF 15 is fine** and SPF 30 will only give you about 4% more protection.
- One the other hand, remembering to **reapply about every 2 hours**, and picking **broad spectrum** formulations are important.



know

calc is FYI

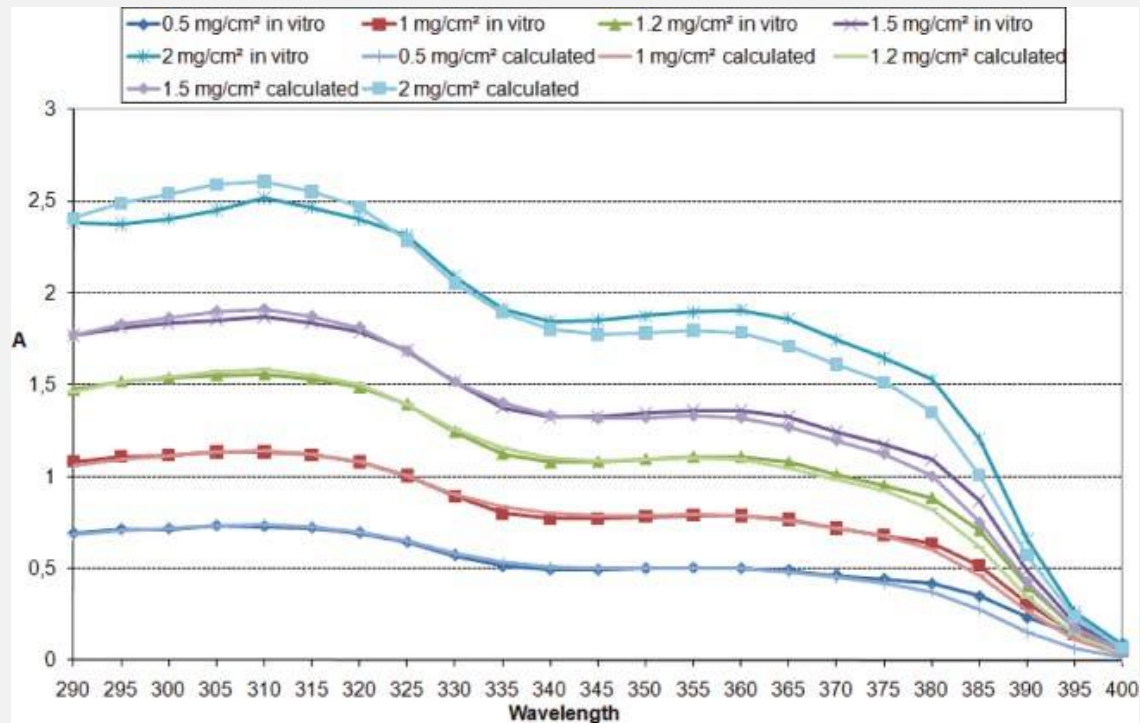
Mathematically, the SPF (or the UPF) is calculated from measured data as

$$\text{SPF} = \frac{\int A(\lambda)E(\lambda)d\lambda}{\int A(\lambda)E(\lambda)/\text{MPF}(\lambda) d\lambda},$$

where $E(\lambda)$ is the solar irradiance spectrum, $A(\lambda)$ the erythral action spectrum, and $\text{MPF}(\lambda)$ the monochromatic protection factor, all functions of the wavelength λ . The MPF is roughly the inverse of the transmittance at a given wavelength.

WHY PROPER AND REAPPLICATION IS RECOMMENDED

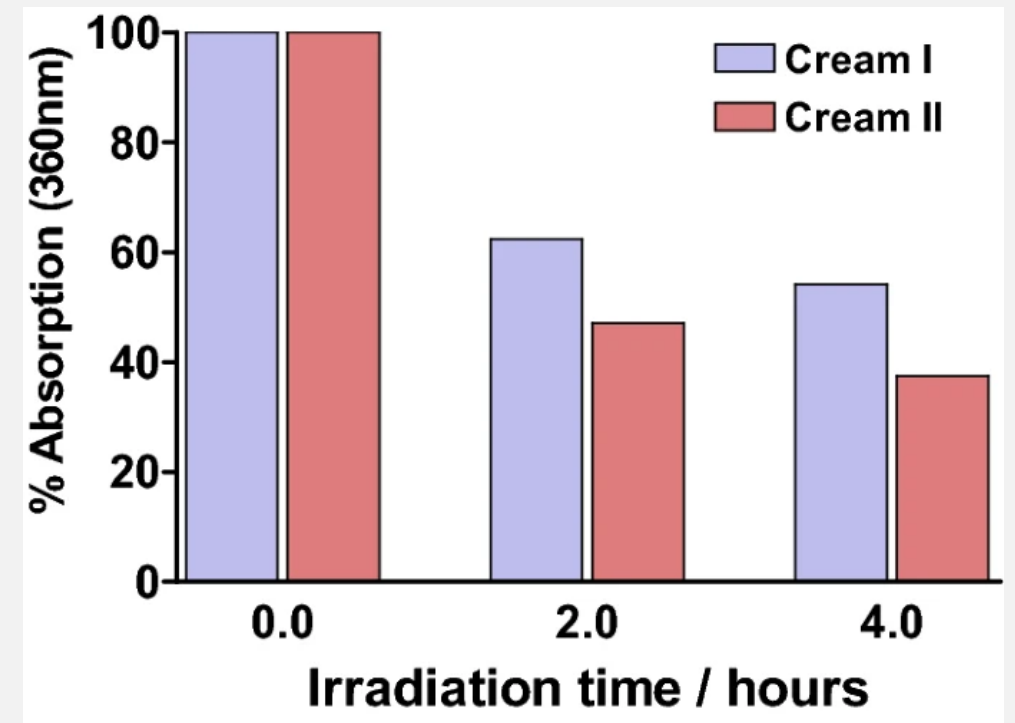
proper application ensures more UV absorbance by the sunscreen



DOI:[10.1039/b9pp00183b](https://doi.org/10.1039/b9pp00183b)

UV absorbance of one sunscreen applied at different average thicknesses

UV absorbance by sunscreens diminishes over time



<https://link.springer.com/article/10.1007/s43630-021-00013-1>

UV absorbance of 2 sunscreens over 4 h (% of 0 h). Cream I: titanium dioxide, ethylhexyl triazone, avobenzone, and octinoxate; Cream II: octyl salicylate, oxybenzone, avobenzone, and octinoxate.

TINTING AND UVA PHOTOPROTECTION WITH IRON OXIDES

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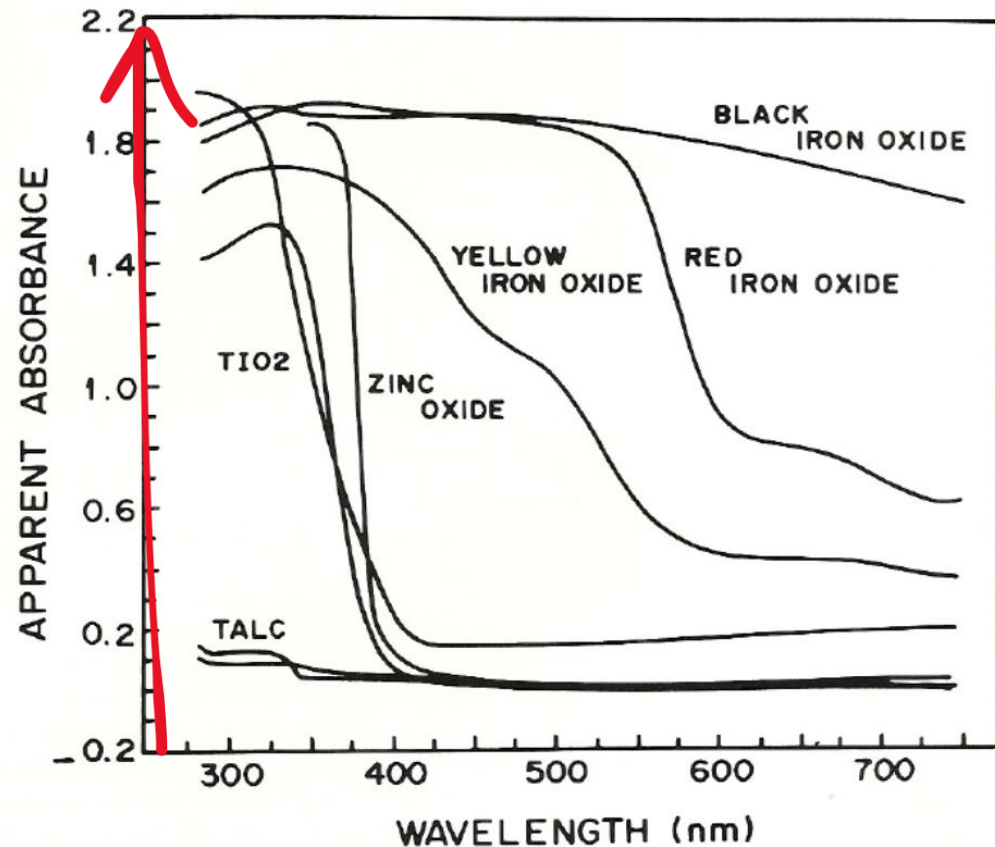
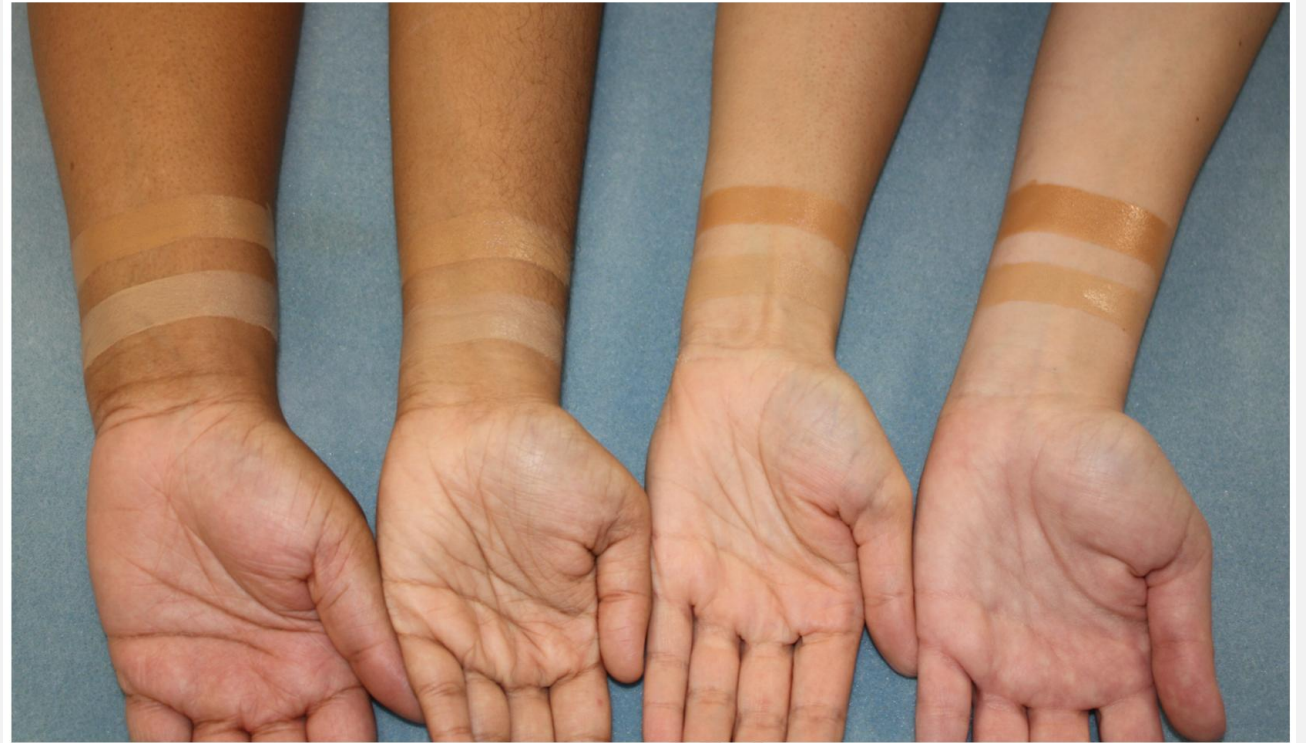


Figure 1. Reflection spectra of physical sunscreen powders. The data plotted shows absorbance vs wavelength. An absorbance of 0.0 corresponds to 100% reflection of the light relative to BaSO_4 ; an absorbance of 1.0 corresponds to 10%; 2.0 corresponds to 1% reflection.

1990 Sayre, et al Journal of the Society of Cosmetic Chemists 41(2):103-109

clinical note: rec. for **melasma**



doi:10.1016/j.jaad.2020.04.079

iron oxides add increased photoprotection in the UVA/vis range AND coloration to sunscreen product



QUICK REVIEW

- What is the product of a Type I psoralen reaction?
- What are the “ABC”s of UV categories, and how much does the atmosphere protect you from each one?
- What active ingredients are probably in a broad-spectrum “chemical” sunscreen?
- What active ingredients are probably in a “tinted mineral” sunscreen?
- How much UVA protection does SPF 80 confer?

QUICK REVIEW

- What is the product of a Type I psoralen reaction?
 - cross-linked DNA adducts
- What are the ABCs of UV wavelength categories, and how much does the atmosphere protect you from each one?
 - Aging (very little), Burning (most, about 90%), Captured (virtually all)
- What active ingredients are probably in a broad-spectrum "chemical" sunscreen?
 - examples: avobenzone, homosalate, octocrylate, etc.
- What active ingredients are probably in a "tinted mineral" sunscreen?
 - TiO_2 , ZnO_2 , iron oxides
- How much UVA protection does SPF 80 confer?
 - haha, who knows?! SPF only applies to *erythema* response, which is caused by UVB exposure. SPF provides no information on UVA damage.

THANKS!

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- please reach out by email with questions or to schedule individual/group office time
 - response will be faster during normal business hours